Bioinformatics

KB8029

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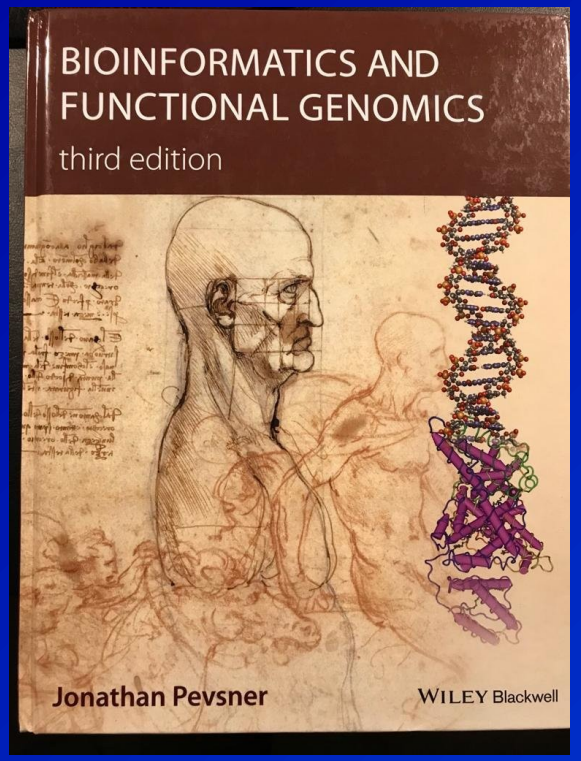
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Sequence Alignment



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Pairwise Sequence Alignment — CHAPTER 3

Multiple Sequence Alignment — CHAPTER 6

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Highlights:

Key terms and definitions

Homology and similarity, orthologs and paralogs

Why sequences needed to be aligned?

Pairwise sequence alignment

Needleman-Wunsch algorithm (global alignment)

Smith-Waterman algorithm (local alignment)

Substitution matrices

PAM family

BLOSUM family

Multiple sequence alignment

Five main approaches

3 Pairwise Sequence Alignment, 69
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Homology and similarity

Homology

Two sequences have a common evolutionary ancestry

Qualitative measure

Orthologs and paralogs

Similarity (identity)

Fraction of similar (identical) residues in related sequences

Quantitative measure

May be used to detect homology

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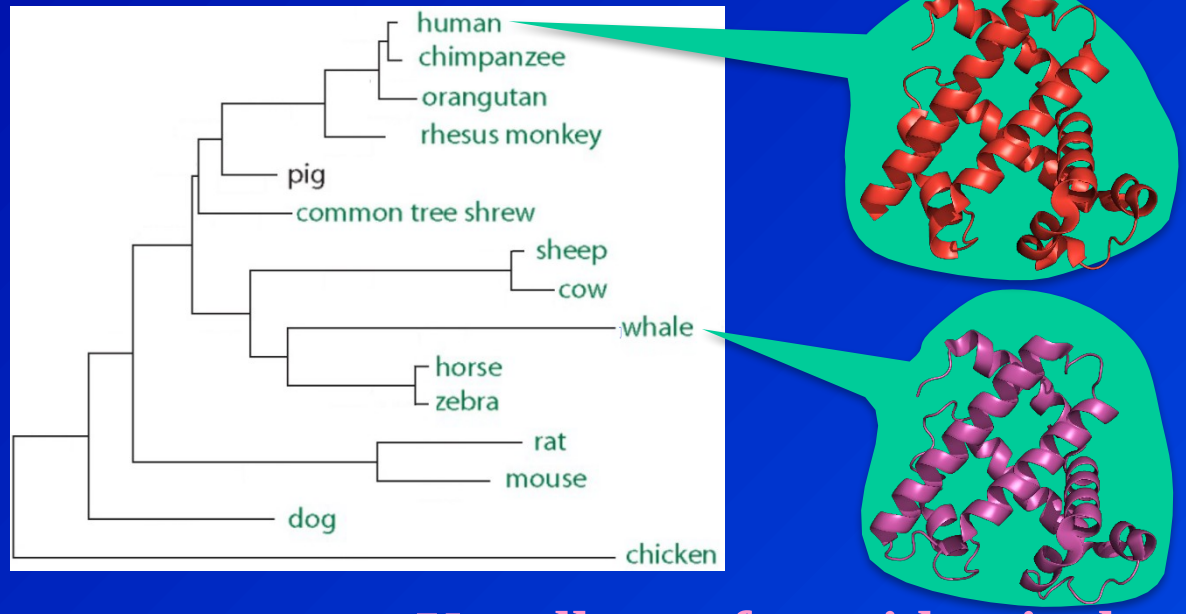
Orthologs and Paralogs

Orthologs

Evolved in *different* species

Gene speciation

Example: myoglobins in different organisms

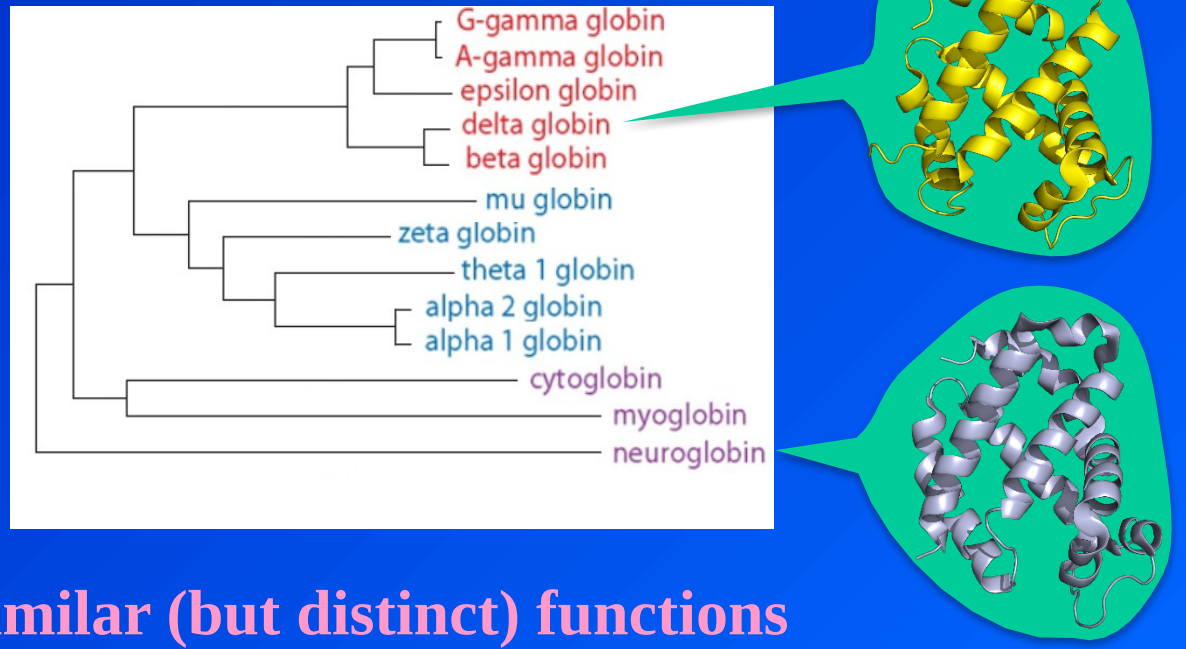


Paralogs

Evolved in *the same* species

Gene duplication

Example: human globins



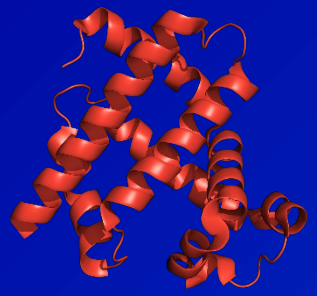
Usually perform identical or similar (but distinct) functions

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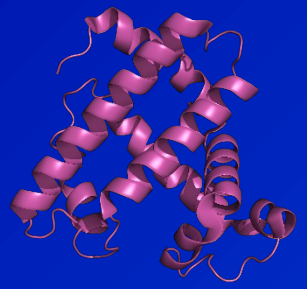
Alignment of sequences

Relatedness of the sequences

Human and sperm whale myoglobins



Human myoglobin, PDB 3rgk, chain A, 154 aa



Whale myoglobin, PDB 101m, chain A, 154 aa

```
Human   1    MGLSDGEWQLVLNVWGKVEADIPGHGQEVLIRLFKGHPETLEKFDKFKHLKSEDEMKASE   60
Whale   1    MVLSEGEWQLVLHVWAKVEADVAGHGQDILIRLFKSHPETLEKFDRFKHLKTEAEMKASE   60

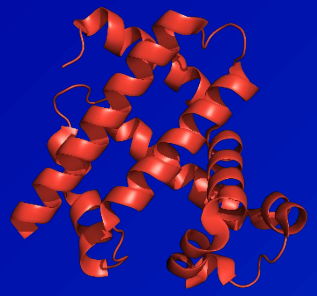
Human   61   DLKKHGATVLTALGGILKKKGHHEAEIKPLAQSHATKHKIPVKYLEFISECIIQVLQSKH   120
Whale   61   DLKKHGVTVLTALGAILKKKGHHEAELKPLAQSHATKHKIPIKYLEFISEAIIHVLHSRH   120

Human   121  PGDFGADAQGAMNKALELFRKDMASNYKELGFQG   154
Whale   121  PGDFGADAQGAMNKALELFRKDIAAKYKELGYQG   154
```

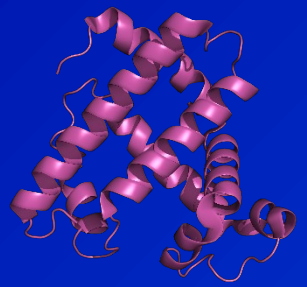
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Human   61   DLKKHGATVLTALGGILKKKGHHEAEIKPLAQSHATKHKIPVKYLEFISECIIQVLQSKH   120
Whale   61   DLKKHGVTVLTALGAILKKKGHHEAELKPLAQSHATKHKIPIKYLEFISEAIIHVLHSRH   120

Human   121  PGDFGADAQGAMNKALELFRKDMASNYKELGFQG   154
Whale   121  PGDFGADAQGAMNKALELFRKDIAAKYKELGYQG   154
```

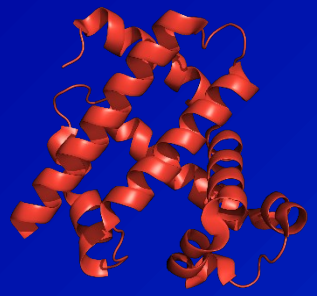
Sequence identity

$$S_{iden} = \frac{N_{blue}}{N_{blue} + N_{red}} \times 100 = \frac{130}{130 + 24} \times 100 = 84.4\%$$

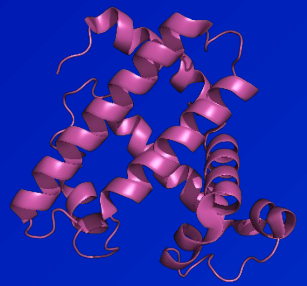
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Alignment of sequences

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Whale myoglobin, P 101m, chain A, 154 aa

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```
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Whale   1    MVLSEGEWQLVLHVWAKVEADVAGHGQDILIRLFKSHPETLEKFDRFKHLKTEAEMKASE   60

Human   61   DLKKHGATVLTALGGILKKKGHHEAEIKPLAQSHATKHKIPVKYLEFISECIIQVLQSKH   120
Whale   61   DLKKHGVTVLTALGAILKKKGHHEAELKPLAQSHATKHKIPIKYLEFISEAIIHVLHSRH   120

Human   121  PGDFGADAQGAMNKALELFRKDMASNYKELGFQG   154
Whale   121  PGDFGADAQGAMNKALELFRKDIAAKYKELGYQG   154
```

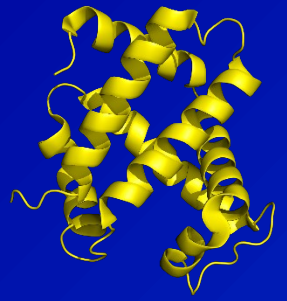
Sequence similarity

$$S_{sim} = \frac{N_{blue} + N_{yellow}}{N_{blue} + N_{yellow} + N_{red}} \times 100 = \frac{130 + 13}{130 + 13 + 11} \times 100 = 92.9\%$$

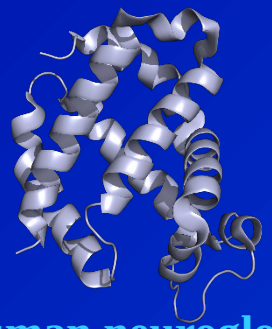
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Alignment of sequences

Relatedness of the sequences



Human delta globin, PDB 1shr, chain A, 147 aa



Human neuroglobin, PDB 1oj6, chain A, 151aa

Human neuro and delta globin

```
Neuro   1   MERPEPELIRQSWRAVSRSPLEHGTVLFARLFALEPDLLPLFQYNCRQFSSPEDCLSSPEFLD  60
Delta   1   MVHLTPEEKTAVNALWGKVNVDAVGGEALGRLLVVYPWTQRFFESFGDLSSPDAVMGNPKVKA  60

Neuro  61   HIRKVMLVIDAAVTNVEDLSSLEEYLASLGRKHRAVGVKLSSFSTVGESLLYMLEKCLGPAFT  120
Delta  61   HGKKVLGAFSDGLAHLDNLKGTFSQLSELHCDKLHVDPENFRLLGNVLVCVLARNFGKEFTPQ  120

Neuro 121   PATRAAWSQLYGAVVQAMSRGWDGE  151
Delta 121   MQAAYQKVVAGVANALAHKYH  147
```

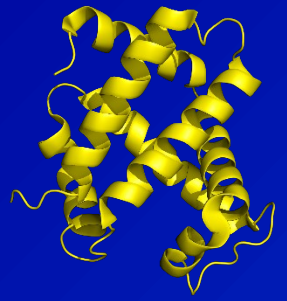
Sequence identity

$$S_{iden} = \frac{N_{blue}}{N_{blue} + N_{red}} \times 100 = \frac{17}{17 + 130} \times 100 = 11.6\%$$

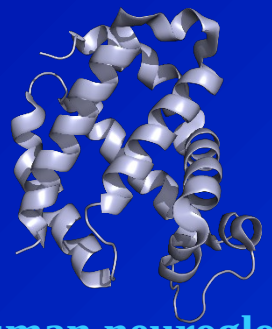
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Alignment of sequences

Relatedness of the sequences



Human delta globin, PDB 1shr, chain A, 147 aa



Human neuroglobin, PDB 1oj6, chain A, 151aa

Human neuro and delta globin

```
Neuro   1   MERPEPELIRQSWRAVSRSPLEHGTVLFARLFALEPDLLPLFQYNCRQFSSPEDCLSSPEFLD  60
Delta   1       MVHLTPEEKTAVNALWGKVNVDAVGGEALGRLLVVYPWTQRFFESFGDLSSPDAVMGNP  56

Neuro  61   HIRKVMLVIDAAVTNVEDLSSLEEYLASLGRKHRAVGVKLSSFSTVGESLLYMLEKCLGPAFT  120
Delta  57   KVKAHGKKVLGAFSDGLAHLDNLKGTFSQLSELHCDKLHVDPENFRLLGNVLVCVLARNFGKE  116

Neuro 121   PATRAAWSQLYGAVVQAMSRGWDGE  151
Delta 117   FTPQMQAAYQKVVAGVANALAHKYH  147
```

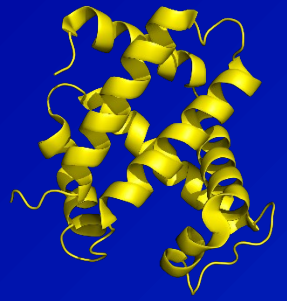
Sequence identity

$$S_{iden} = \frac{N_{blue}}{N_{blue} + N_{red}} \times 100 = \frac{6}{6 + 141} \times 100 = 4.1\%$$

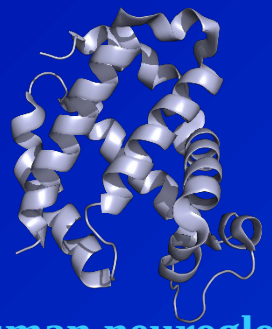
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Alignment of sequences

Relatedness of the sequences



Human delta globin, PDB 1shr, chain A, 147 aa



Human neuroglobin, PDB 1oj6, chain A, 151aa

Human neuro and delta globin

```
Neuro    5   EPELIRQSWRAVSRSPLEHGTVLFARLFALEPDLLPLFQYNCRQFSSPEDCLSSPEFLDH  64
Delta    8   EKTAVNALWGKVNVDAV--GGEALGRLLVVYPWTQRFFE-SFGDLSSPDAVMGNPKVKAH  64

Neuro   65   IRKVMLVIDAAVTNVEDLSSLEEYLASLGRKHRAVGVKLSSFSTVGESLLYMLEKCLGPA  124
Delta   65   GKKVLGAFSDGLAHLDNLKGTFSQLSEL--HCDKLHVDPENFRLLGNVLVCVLARNFGKE  122

Neuro  125   FTPATRAAWSQLYGAVVQAMSRGW  148
Delta  123   FTPQMQAAYQKVVAGVANALAHKY  146
```

Sequence identity

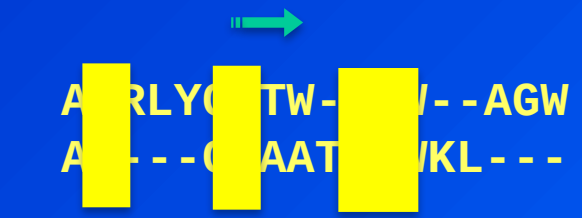
$$s_{iden} = \frac{N_{blue}}{N_{blue} + N_{red}} \times 100 = \frac{31}{31 + 113} \times 100 = 21.5\%$$

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Alignment of sequences

Rearranging sequences to infer:

ADRLYGGTWQQAGW
ADGGAATQQWKL



Why?

regions of similarity and evolutionary deletion/insertion

regions of evolutionary deletion/insertion

possibly homology

Maximize value of an alignment score

How?

Aligning two residues

Exact matches

Some are more equal than others (certain amino acid pairs are preferable)

Introducing/extending gaps

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Alignment of sequences

”Brut-force” approach ?

calculate scores for all N possible alignments

Test case (two 3 aa chains)

AGN

ADN

No gaps (1 possibility)	One gap (12 possibilities)	Two gaps (30 possibilities)	Three gaps (20 possibilities)
AGN ADN	-AGN A-DN -AGN A-DN -AGN AND- A-GN -AND A-GN AN-D A-GN AND- AG-N -AND AG-N A-ND AG-N AND- AGN- -AND AGN- A-ND AGN- AN-D	--AGN AD--N .. AGN-- A--DN	---AGN ADN--- .. -A–G-N A-D-N-

$\mathcal{N}$ = 1 + 12 + 30 + 20 = 63

Two proteins with *n* amino acids

$$N = \frac{2n!}{(n!)^2} \approx \frac{2^{2n}}{\sqrt{\pi n}}$$

$n=100;\ \mathcal{N} \approx 10^{59}$

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Alignment of sequences

”Brut-force” approach ?

calculate scores for all N possible alignments

Test case (two 3 aa chains)

AGN

ADN

Prohibitive!

No gaps (1 possibility)

One gaps (12 possibilities)

Two gaps (30 possibilities)

Three gaps (20 possibilities)

$\mathcal{N}=$ AGN / ADN + (-AGN / A-DN) (-AGN / A-DN) (-AGN / AND-) (A-GN / -AND) (A-GN / AN-D) (A-GN / AND-) (AG-N / -AND) (AG-N / A-ND) (AG-N / AND-) (AGN- / -AND) (AGN- / A-ND) (AGN- / AN-D) + (- -AGN / AD- -N) .. (AGN- - / A- -DN) + (- - -AGN / ADN- - -) .. (-A–G-N / A-D-N-) $= 63$

Two proteins with *n* amino acids

$$N = \frac{2n!}{(n!)^2} \approx \frac{2^{2n}}{\sqrt{\pi n}}$$

$n=100;\ \mathcal{N} \approx 10^{59}$

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Alignment of sequences

Computational algorithms

Dotplot protocol

Dynamical programming

Needleman-Wunsch global alignment

Smith-Waterman local alignment

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Dotplot protocol

Simplest technique

	i	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	2
j		S	E	Q	U	E	N	C	E	A	N	A	L	Y	S	I	S	P	R	I	M	E	R
1	A									■		■											
2	N						■				■												
3	A									■		■											
4	L												■										
5	Y													■									
6	Z																						
7	E		■			■			■													■	
8	S	■													■		■						
9	E		■			■			■													■	
10	Q			■																			
11	U				■																		
12	E		■			■			■													■	
13	N						■				■												
14	C							■															
15	E		■			■			■													■	
16	S	■													■		■						

(*i*, *j*)

Still used in alignment output visualisation

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Dynamical programming

Sequence 1: x = {AGCNVRQCLCRPM} N_1 = 13 residues

Sequence 2: y = {AVCVNRCKCRGP} N_2 = 12 residues

Step 1

Construct a DP matrix, V=[(N_1+1)×(N_2+1)] using

Gap opening score (g = 1)

Scores for aligning two residues, matrix s

s(A,A)= 2
s(A,G)= 0
s(A,C)=-1
s(A,V)=-1
s(A,C)=-1
s(V,G)=-2
…

Step 2

Construct an optinal alignment by the traceback in the DP matrix V

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Dynamical programming

Needleman-Wunsch global alignment algorithm

Sequence 1: x = {AGCNVRQCLCRPM} N_1 = 13 residues

Sequence 2: y = {AVCVNRCKCRGP} N_2 = 12 residues

DP matrix, V, with gap opening penalty *g*=1 and simple scoring matrix *S*

	0	A	G	C	…	M
0	0	-g	-2g	-3g	…	-13g
A	-g					
V	-2g					
C	-3g					
…	…					
P	-12g					

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Dynamical programming

Needleman-Wunsch global alignment algorithm

Sequence 1: x = {AGCNVRQCLCRPM} N_1 = 13 residues

Sequence 2: y = {AVCVNRCKCRGP} N_2 = 12 residues

DP matrix, V, with gap opening penalty g=1 and simple scoring matrix S

	0	A	G	C	…	M
0	0	-1	-2	-3	…	-13
A	-1					
V	-2					
C	-3					
…	…					
P	-12					

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DP matrix, V, with gap opening penalty g=1 and simple scoring matrix S

	0	A	G	C	…	M
0	0	-g	-2g	-3g	…	-13g
A	-g					
V	-2g					
C	-3g					
…	…					
P	-12g					

$$v_{i,j} = \max \begin{cases} v_{i-1,j-1} + s(x_i, y_j) \\ v_{i-1,j} - g \\ v_{i,j-1} - g \end{cases}$$

Scoring matrix:

s(A,A)= 2

s(A,G)= 0

s(A,C)=-1

s(V,A)=-1

s(C,A)=-1

…

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Dynamical programming

Needleman-Wunsch global alignment algorithm

Sequence 1: x = {AGCNVRQCLCRPM} N_1 = 13 residues

Sequence 2: y = {AVCVNRCKCRGP} N_2 = 12 residues

DP matrix, V, with gap opening penalty g=1 and simple scoring matrix S

	0	A	G	C	…	M
0	0	-1	-2	-3	…	-13
A	-1	*				
V	-2					
C	-3					
…	…					
P	-12					

$$v_{1,1} = \max \begin{cases} v_{0,0} + s(x_1, y_1) \\ v_{0,1} - g \\ v_{1,0} - g \end{cases}$$

s(A,A)= 2 s(A,G)= 0

s(A,C)=-1 s(A,V)=-1

s(A,C)=-1 s(V,G)=-2

…

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Dynamical programming

Needleman-Wunsch global alignment algorithm

Sequence 1: x = {AGCNVRQCLCRPM} N_1 = 13 residues

Sequence 2: y = {AVCVNRCKCRGP} N_2 = 12 residues

DP matrix, V, with gap opening penalty g=1 and simple scoring matrix S

	0	A	G	C	…	M
0	0	-1	-2	-3	…	-13
A	-1	2				
V	-2					
C	-3					
…	…					
P	-12					

$$v_{1,1} = \max \begin{cases} v_{0,0} + s(A,A) \\ v_{0,1} - g \\ v_{1,0} - g \end{cases} = \max \begin{cases} 2 \\ -2 \\ -2 \end{cases} = 2$$

s(A,A)= 2 s(A,G)= 0

s(A,C)=-1 s(A,V)=-1

s(A,C)=-1 s(V,G)=-2

…

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Dynamical programming

Needleman-Wunsch global alignment algorithm

Sequence 1: x = {AGCNVRQCLCRPM} N_1 = 13 residues

Sequence 2: y = {AVCVNRCKCRGP} N_2 = 12 residues

DP matrix, V, with gap opening penalty *g*=1 and simple scoring matrix *S*

	0	A	G	C	…	M
0	0	-1	-2	-3	…	-13
A	-1	2				
V	-2					
C	-3					
…	…					
P	-12					



s(A,A)= 2 s(A,G)= 0

s(A,C)=-1 s(A,V)=-1

s(A,C)=-1 s(V,G)=-2

…

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Dynamical programming

Needleman-Wunsch global alignment algorithm

Sequence 1: x = {AGCNVRQCLCRPM} N_1 = 13 residues

Sequence 2: y = {AVCVNRCKCRGP} N_2 = 12 residues

DP matrix, V, with gap opening penalty *g*=1 and simple scoring matrix *S*

	0	A	G	C	…	M
0	0	-1	-2	-3	…	-13
A	-1	2	1			
V	-2					
C	-3					
…	…					
P	-12					



s(A,A)= 2 s(A,G)= 0

s(A,C)=-1 s(A,V)=-1

s(A,C)=-1 s(V,G)=-2

…

Remember the path

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Dynamical programming

Needleman-Wunsch global alignment algorithm

Sequence 1: x = {AGCNVRQCLCRPM} N_1 = 13 residues

Sequence 2: y = {AVCVNRCKCRGP} N_2 = 12 residues

DP matrix, V, with gap opening penalty g=1 and simple scoring matrix S

	0	A	G	C	…	M
0	0	-1	-2	-3	…	-13
A	-1	2	1	0		
V	-2	1	0			
C	-3	0				
…	…					
P	-12					

s(A,A)= 2

s(A,G)= 0

s(A,C)=-1

s(A,V)=-1

s(A,C)=-1

s(V,G)=-2

…

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Dynamical programming

Needleman-Wunsch global alignment algorithm

Traceback procedure

	0	A	G	C	N	V	R	Q	C	L	C	R	P	M
0	0	-1	-2	-3	-4	-5	-6	-7	-8	-9	-10	-11	-12	-13
A	-1	2	1	0	-1	-2	-3	-4	-5	-6	-7	-8	-9	-10
V	-2	1	0	0	-1	1	0	-1	-2	-3	-4	-5	-6	-7
C	-3	0	0	3	2	1	0	-1	1	0	-1	-2	-3	-4
V	-4	-1	-1	2	2	4	3	2	1	0	-1	-2	-3	-4
N	-5	-2	-2	1	4	3	3	2	1	0	-1	-2	-3	-4
R	-6	-3	-3	0	3	3	5	4	3	2	1	1	0	-1
C	-7	-4	-4	-1	2	2	4	4	6	5	4	3	2	1
K	-8	-5	-5	-2	1	1	3	3	5	5	4	3	2	1
C	-9	-6	-6	-3	0	0	2	2	5	4	7	6	5	4
R	-10	-7	-7	-4	-1	-1	2	1	4	4	6	11	8	7
G	-11	-8	-8	-5	-2	-2	1	0	3	3	5	8	8	7
P	-12	-9	-9	-6	-3	-3	0	-1	2	2	4	7	10	9

Start

Stockholms universitet

Dynamical programming

Needleman-Wunsch global alignment algorithm

Traceback procedure

	0	A	G	C	N	V	R	Q	C	L	C	R	P	M
0	0	-1	-2	-3	-4	-5	-6	-7	-8	-9	-10	-11	-12	-13
A	-1	2	1	0	-1	-2	-3	-4	-5	-6	-7	-8	-9	-10
V	-2	1	0	0	-1	1	0	-1	-2	-3	-4	-5	-6	-7
C	-3	0	0	3	2	1	0	-1	1	0	-1	-2	-3	-4
V	-4	-1	-1	2	2	4	3	2	1	0	-1	-2	-3	-4
N	-5	-2	-2	1	4	3	3	2	1	0	-1	-2	-3	-4
R	-6	-3	-3	0	3	3	5	4	3	2	1	1	0	-1
C	-7	-4	-4	-1	2	2	4	4	6	5	4	3	2	1
K	-8	-5	-5	-2	1	1	3	3	5	5	4	3	2	1
C	-9	-6	-6	-3	0	0	2	2	5	4	7	6	5	4
R	-10	-7	-7	-4	-1	-1	2	1	4	4	6	11	8	7
G	-11	-8	-8	-5	-2	-2	1	0	3	3	5	8	8	7
P	-12	-9	-9	-6	-3	-3	0	-1	2	2	4	7	10	9

Start

M

-

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Dynamical programming

Needleman-Wunsch global alignment algorithm

Traceback procedure

	0	A	G	C	N	V	R	Q	C	L	C	R	P	M
0	0	-1	-2	-3	-4	-5	-6	-7	-8	-9	-10	-11	-12	-13
A	-1	2	1	0	-1	-2	-3	-4	-5	-6	-7	-8	-9	-10
V	-2	1	0	0	-1	1	0	-1	-2	-3	-4	-5	-6	-7
C	-3	0	0	3	2	1	0	-1	1	0	-1	-2	-3	-4
V	-4	-1	-1	2	2	4	3	2	1	0	-1	-2	-3	-4
N	-5	-2	-2	1	4	3	3	2	1	0	-1	-2	-3	-4
R	-6	-3	-3	0	3	3	5	4	3	2	1	1	0	-1
C	-7	-4	-4	-1	2	2	4	4	6	5	4	3	2	1
K	-8	-5	-5	-2	1	1	3	3	5	5	4	3	2	1
C	-9	-6	-6	-3	0	0	2	2	5	4	7	6	5	4
R	-10	-7	-7	-4	-1	-1	2	1	4	4	6	11	8	7
G	-11	-8	-8	-5	-2	-2	1	0	3	3	5	8	8	7
P	-12	-9	-9	-6	-3	-3	0	-1	2	2	4	7	10	9

Start

PM

P-

Stockholms universitet

Dynamical programming

Needleman-Wunsch global alignment algorithm

Traceback procedure

	0	A	G	C	N	V	R	Q	C	L	C	R	P	M
0	0	-1	-2	-3	-4	-5	-6	-7	-8	-9	-10	-11	-12	-13
A	-1	2	1	0	-1	-2	-3	-4	-5	-6	-7	-8	-9	-10
V	-2	1	0	0	-1	1	0	-1	-2	-3	-4	-5	-6	-7
C	-3	0	0	3	2	1	0	-1	1	0	-1	-2	-3	-4
V	-4	-1	-1	2	2	4	3	2	1	0	-1	-2	-3	-4
N	-5	-2	-2	1	4	3	3	2	1	0	-1	-2	-3	-4
R	-6	-3	-3	0	3	3	5	4	3	2	1	1	0	-1
C	-7	-4	-4	-1	2	2	4	4	6	5	4	3	2	1
K	-8	-5	-5	-2	1	1	3	3	5	5	4	3	2	1
C	-9	-6	-6	-3	0	0	2	2	5	4	7	6	5	4
R	-10	-7	-7	-4	-1	-1	2	1	4	4	6	11	8	7
G	-11	-8	-8	-5	-2	-2	1	0	3	3	5	8	8	7
P	-12	-9	-9	-6	-3	-3	0	-1	2	2	4	7	10	9

Start

-PM

GP-

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Dynamical programming

Needleman-Wunsch global alignment algorithm

Traceback procedure

End

	0	A	G	C	N	V	R	Q	C	L	C	R	P	M
0	0	-1	-2	-3	-4	-5	-6	-7	-8	-9	-10	-11	-12	-13
A	-1	2	1	0	-1	-2	-3	-4	-5	-6	-7	-8	-9	-10
V	-2	1	0	0	-1	1	0	-1	-2	-3	-4	-5	-6	-7
C	-3	0	0	3	2	1	0	-1	1	0	-1	-2	-3	-4
V	-4	-1	-1	2	2	4	3	2	1	0	-1	-2	-3	-4
N	-5	-2	-2	1	4	3	3	2	1	0	-1	-2	-3	-4
R	-6	-3	-3	0	3	3	5	4	3	2	1	1	0	-1
C	-7	-4	-4	-1	2	2	4	4	6	5	4	3	2	1
K	-8	-5	-5	-2	1	1	3	3	5	5	4	3	2	1
C	-9	-6	-6	-3	0	0	2	2	5	4	7	6	5	4
R	-10	-7	-7	-4	-1	-1	2	1	4	4	6	11	8	7
G	-11	-8	-8	-5	-2	-2	1	0	3	3	5	8	8	7
P	-12	-9	-9	-6	-3	-3	0	-1	2	2	4	7	10	9

Start

AGCNV-RQCLCR-PM
AVC-VNR-CKCRGP-

Best alignment score

Stockholms universitet

Dynamical programming

Needleman-Wunsch global alignment algorithm

Traceback procedure

End

	0	A	G	C	N	V	R	Q	C	L	C	R	P	M
0	0	-1	-2	-3	-4	-5	-6	-7	-8	-9	-10	-11	-12	-13
A	-1	2	1	0	-1	-2	-3	-4	-5	-6	-7	-8	-9	-10
V	-2	1	1	0	-1	1	0	-1	-2	-3	-4	-5	-6	-7
C	-3	0	0	3	2	1	0	-1	1	0	-1	-2	-3	-4
V	-4	-1	-1	2	2	4	3	2	1	0	-1	-2	-3	-4
N	-5	-2	-2	1	4	3	3	2	1	0	-1	-2	-3	-4
R	-6	-3	-3	0	3	3	5	4	3	2	1	1	0	-1
C	-7	-4	-4	-1	2	2	4	4	6	5	4	3	2	1
K	-8	-5	-5	-2	1	1	3	3	5	5	4	3	2	1
C	-9	-6	-6	-3	0	0	2	2	5	4	7	6	5	4
R	-10	-7	-7	-4	-1	-1	2	1	4	4	6	11	8	7
G	-11	-8	-8	-5	-2	-2	1	0	3	3	5	8	8	7
P	-12	-9	-9	-6	-3	-3	0	-1	2	2	4	7	10	9

Start

AGCNV-RQCLCR-PM
AVC-VNR-CKCRGP-

AGC-NVRQCLCR-PM
AVCVN-R-CKCRGP-

Best alignment score

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Dynamical programming

Smith-Waterman local alignment algorithm

Sequence 1: x = {AGCNVRQCLCRPM} N_1 = 13 residues

Sequence 2: y = {AVCVNRCKCRGP} N_2 = 12 residues

DP matrix, V, with gap opening penalty g=1 and simple scoring matrix S

	0	A	G	C	…	M
0	0	0	0	0	…	0
A	0					
V	0					
C	0					
…	…					
P	0					



$$v_{i,j} = \max \begin{cases} v_{i-1,j-1} + s(x_i, y_j) \\ v_{i-1,j} - g \\ v_{i,j-1} - g \\ 0 \end{cases}$$

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Dynamical programming

Smith-Waterman local alignment algorithm

Sequence 1: x = {AGCNVRQCLCRPM} N_1 = 13 residues

Sequence 2: y = {AVCVNRCKCRGP} N_2 = 12 residues

DP matrix, V, with gap opening penalty *g*=1 and simple scoring matrix *S*

	0	A	G	C	…	M
0	0	0	0	0	…	0
A	0	2	1	0		
V	0	1	0			
C	0	0				
…	…					
P	0					

s(A,A)= 2

s(A,G)= 0

s(A,C)=-1

s(A,V)=-1

s(A,C)=-1

s(V,G)=-2

…

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Dynamical programming

Smith-Waterman local alignment algorithm

Traceback procedure

	0	A	G	C	N	V	R	Q	C	L	C	R	P	M
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
A	0	2	1	0	0	0	0	0	0	0	0	0	0	0
V	0	1	0	0	0	1	0	0	0	0	0	0	0	0
C	0	0	0	3	2	1	0	0	1	0	0	0	0	0
V	0	0	0	2	2	4	3	2	1	0	0	0	0	0
N	0	0	0	1	4	3	3	2	1	0	0	0	0	0
R	0	0	0	0	3	3	5	4	3	2	1	1	0	0
C	0	0	0	0	2	2	4	4	6	5	4	3	2	1
K	0	0	0	0	1	1	3	3	5	5	4	3	2	1
C	0	0	0	0	0	0	2	2	5	4	7	6	5	4
R	0	0	0	0	0	0	2	1	4	4	6	11	8	7
G	0	0	0	0	0	0	1	0	3	3	5	8	8	7
P	0	0	0	0	0	0	0	0	2	2	4	7	10	9

Start

Stockholms universitet

Dynamical programming

Smith-Waterman local alignment algorithm

Traceback procedure

End

	0	A	G	C	N	V	R	Q	C	L	C	R	P	M
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
A	0	2	1	0	0	0	0	0	0	0	0	0	0	0
V	-2	1	0	0	0	1	0	0	0	0	0	0	0	0
C	0	0	0	3	2	1	0	0	1	0	0	0	0	0
V	0	0	0	2	2	4	3	2	1	0	0	0	0	0
N	0	0	0	1	4	3	3	2	1	0	0	0	0	0
R	0	0	0	0	3	3	5	4	3	2	1	1	0	0
C	0	0	0	0	2	2	4	4	6	5	4	3	2	1
K	0	0	0	0	1	1	3	3	5	5	4	3	2	1
C	0	0	0	0	0	0	2	2	5	4	7	6	5	4
R	0	0	0	0	0	0	2	1	4	4	6	11	8	7
G	0	0	0	0	0	0	1	0	3	3	5	8	8	7
P	0	0	0	0	0	0	0	0	2	2	4	7	10	9

Start

CNV-RQCLCR
C-VNR-CKCR

Best alignment score

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What is needed to align?

Score of matching two amino acids $s(x_i, y_j)$

Simple substitution matrices (20 × 20)

Position-specific scoring matrix ($N_1 \times N_2$)

Gap penalty function, $g(n)$

Linear function

$$g(n) = n \times g_{op}, \quad g_{op} \sim s(x_i, x_j)$$

Affine gap penalty function

$$g(n) = g_{op} + (n-1) \times g_{ext}, \quad g_{ext} << g_{op}$$

Position-dependent gap penalty function

$$g(n,i) = \left\{ g_{op} + (n-1) \times g_{ext} \right\} \times \exp(n_i), \quad n_i = 0 \text{ or } 1$$



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Substitution matrices

PAM (Point Accepted Mutations) family

M.D. Dayhoff, R.M. Schwartz, and B.C. Orcut (1978), In *Atlas of Protein Sequences and Structures*, pp. 345 - 352

BLOSUM (BLOck SUbstitution Matrix) family

S. Henikoff and J.G.Henikoff (1992), Amino acids substitution matrices from protein blocks, PNAS, 89, 10915-10919

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Substitution matrices

PAM (Point Accepted Mutations) family

Needleman-Wunsch global alignments of 34 protein superfamilies

Source: Atlas of Protein Sequence and Structure, ~800 sequences at the time

Well conserved families

Not conserved families

Non-conserved positions

Highly variable positions

Conserved positions

Glyceraldehyde-3-phosphate dehydrogenase proteins

```
P04406  Human          164 IHDNFGIVEGLMTTVHAITATQKTVDGPSGKLWRDGRGALQNII 207
G3R288  Chimpanzee     164 IHDNFGIVEGLMTTVHAITATQKTVDGPSGKLWRDGRGALQNII 207
Q28259  Dog            162 IHDHFGIVEGLMTTVHAITATQKTVDGPSGKMWRDGRGAAQNII 205
D3YU05  Mouse          168 IHDNFGIMEGLMTTVHAITATQKTVDGPSGKLWRDGRGAAQNII 211
P04797  Rat            162 IHDNFGIVEGLMTTVHAITATQKTVDGPSGKLWRDGRGAAQNII 205
P00356  Chicken        162 IHDNFGIVEGLMTTVHAITATQKTVDGPSGKLWRDGRGAAQNII 205
Q01597  Fruit fly      161 INDNFEIVEVLMTTVHATTATQKTVDGPSGKLWRDGRGACQNII 204
Q7Q1U8  Mosquito       161 INDNFGILEGLMTTVHATTATQKTVDGPSGKLWRDGRGAAQNII 204
P17329  C.elegans      170 INDNFGIIEGLMTTVHAVTATQKTVDGPSGKLWRDGRGAGQNII 213
O43026  Fission yeast  164 INDTFGIEEGLMTTVHATTATQKTVDGPSKKDWRGGRGASANII 207
P00359  Baker's yeast  162 INDAFGIEEGLMTTVHSLTATQKTVDGPSHKDWRGGRTASGNII 205
P84998  Fungus         161 INDEFGIDEALMTTVHSITATQKTVDGPSHKDWRGGRTASGNII 204
Q0J8A4  Rice           166 INDEFGIEEGLMTTVHSITATQKTVDGPSHKDWRGGRTASGNII 209
```

Kappa-casein precursor proteins

```
P07498  Human    111 PNLHPSFIAIPPKKIQDKIIPTTINTIATVEPT--PAPAT------EPTVDSVVTPEAFS 162
P82187  Horse    111 PCPHPSFIAIPPKKLQEITVIPKINTIATVEPT--PIPTP------EPTVNNAVIPDASS 162
P11841  Pig      111 PRPHASFIAIPPKKNQDKTAIPAINSIATVEPT--IVPATEPIVNAEPIVNAVVTPEASS 168
P02669  Sheep    120 PHPHLSFMAIPPKKDQDKTEIPAINTIASAEPTVHSTPTT------EAVVNAVDNPEASS 173
P33618  Rabbit   112 S--HPFFMAILPNKMQDKAVTPTTNTIAAVEPT--PIPTT------EPVVSTEVIAEASP 161
P02668  Cattle   119 PHPHLSFMAIPPKKNQDKTEIPTINTIASGEPT--STPTT------EAVESTVATLEDSP 171
P06796  Mouse    112 AIPNPSFLAMPTNENQDNTAIPTIDPITPIVST--PVPTM------ESIVNTVANPEAST 163
```

M.D. Dayhoff, R.M. Schwartz, and B.C. Orcut (1978), In *Atlas of Protein Sequences and Structures*, pp. 345 - 352

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Substitution matrices

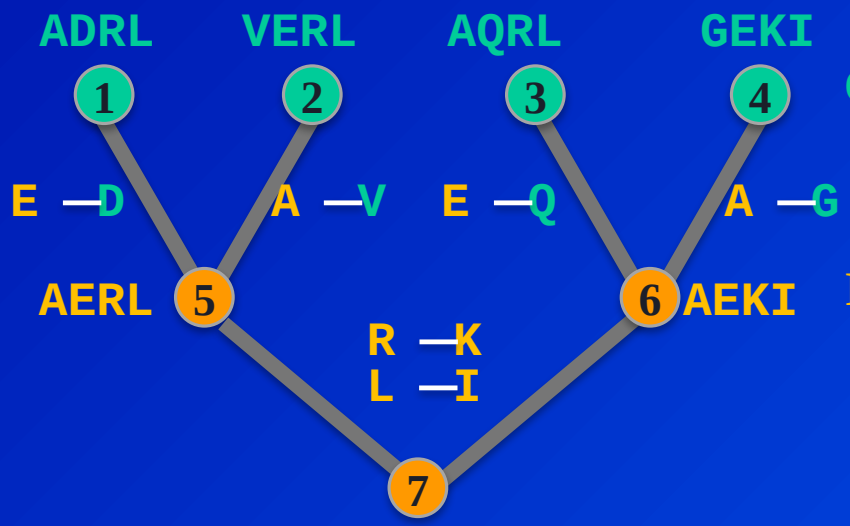
PAM (Point Accepted Mutations) family

organized sequences in phylogenetic trees

predicted sequences of ancestors

counted substitutions between observed and ancestor sequences

Phylogenetic tree



Observed sequences

Predicted sequences

Exemplar substitution matrix, A

	A	G	I	L	V	Q	K	R	D	E
A		1			1					
G	1									
I				1						
L			1							
V	1									
Q										1
K								1		
R							1			
D										1
E						1			1	

M.D. Dayhoff, R.M. Schwartz, and B.C. Orcut (1978), In *Atlas of Protein Sequences and Structures*, pp

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Substitution matrices

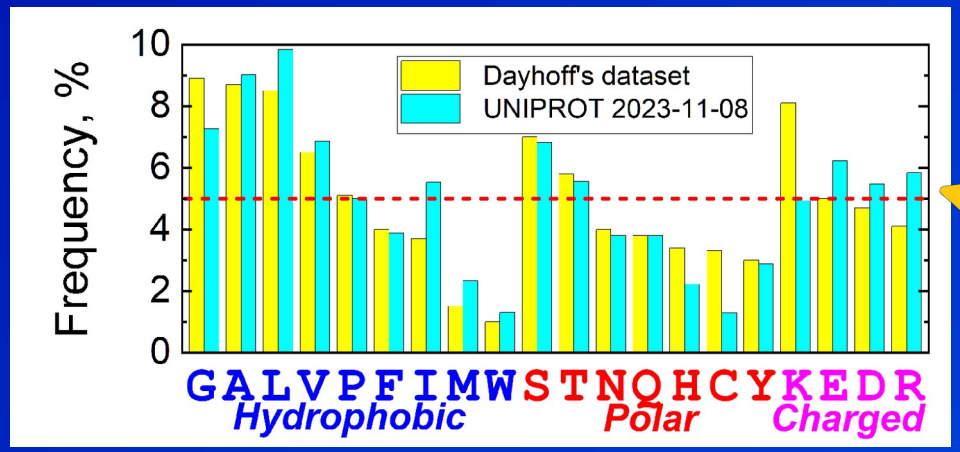
PAM (Point Accepted Mutations) family

frequency of amino acids

$$f(x) = \frac{n(x)}{\sum_{\{x\}} n(x)}$$

Number of times residue x is observed in a dataset

Total number of residues in the dataset



Random probability, 1/20

M.D. Dayhoff, R.M. Schwartz, and B.C. Orcut (1978), In *Atlas of Protein Sequences and Structures*, pp. 345 - 352

Stockholms universitet

Substitution matrices

PAM (Point Accepted Mutations) family

relative mutability of amino acids

$$\mu(x) = \frac{a(x)}{n(x)} = \frac{\sum_{\{y \neq x\}} a(x \rightarrow y)}{n(x)}$$

Number of times residue x is observed to be replaced by a residue y in a dataset

Number of times residue x appears in the dataset

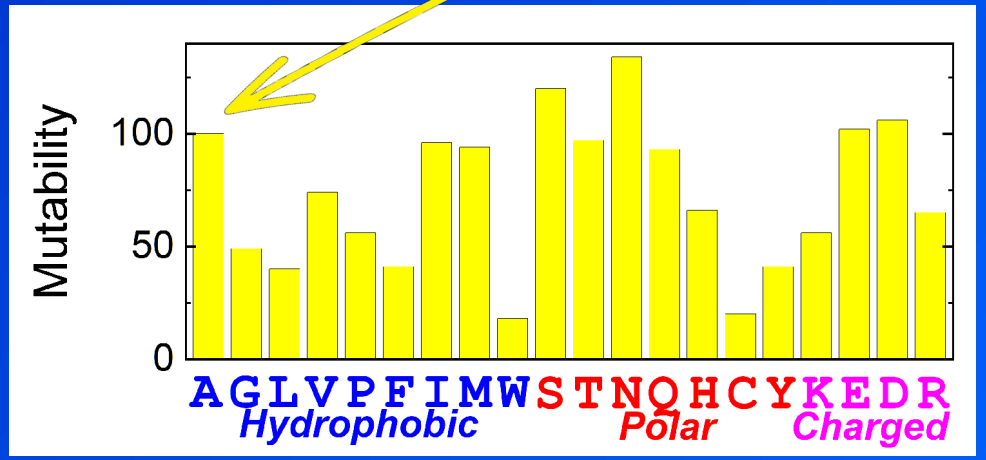
Exemplar substitution matrix, A

	A	G	I	L	V	Q	K	R	D	E
A		1			1					
G	1									
I				1						
L			1							
V	1									
Q										1
K								1		
R							1			
D										1

a(A→ G)

a(A)=a(A→ G)+ +a(A→ V)

Arbitrary set to 100



ayhoff, R.M. Schwartz, and B.C. Orcut (1978), In *Atlas of Protein Sequences and Structures*, pp. 345 - 352

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Substitution matrices

PAM (Point Accepted Mutations) family

PAM1 mutation matrix, M

Only sequences in the trees with no more then 1% of amino acids changed

Non-diagonal elements ($x \neq y$)

Mutability of residue

Element of the substitution matrix

$$m(x \rightarrow y) = \frac{\lambda \mu(x) a(x \rightarrow y)}{\sum_{\{y \neq x\}} a(x \rightarrow y)} = \frac{\lambda a(x \rightarrow y)}{n(x)}$$

Proportionality constant

Number of times residue x is observed in a dataset

Diagonal elements ($x = y$)

$$m(x \rightarrow x) = 1 - \sum_{\{y \neq x\}} m(x \rightarrow y) = 1 - \lambda \mu(x)$$

Proportionality constant λ

$$\frac{\sum_{\{x\}} n(x) m(x \rightarrow x)}{\sum_{\{x\}} n(x)} = 1 - \lambda \sum_{\{x\}} f(x) \mu(x)$$

Fraction of conserved residues, 0.99 for PAM1

M.D. Dayhoff, R.M. Schwartz, and B.C. Orcut (1978), In *Atlas of Protein Sequences and Structures*, pp. 345 - 352

Stockholms universitet

Substitution matrices

PAM (Point Accepted Mutations) family

Replacement amino acids

Properties of PAM1 mutation matrix

Probability (in %) that Ala will not mutated at evolutionary distance 1 PAM

Original amino acids

	Ala	Arg	Asn	Asp	Cys	Gln	Glu	Gly
Ala	98.7	0.0	0.1	0.1	0.0	0.1	0.2	0.2
Arg	0.0	99.1	0.0	0.0	0.0	0.1	0.0	0.0
Asn	0.0	0.0	98.2	0.4	0.0	0.0	0.1	0.1
Asp	0.1	0.0	0.4	98.6	0.0	0.1	0.5	0.1
Cys	0.0	0.0	0.0	0.0	99.7	0.0	0.0	0.0
Gln	0.0	0.1	0.0	0.1	0.0	98.8	0.3	0.0
Glu	0.1	0.0	0.1	0.6	0.0	0.4	98.7	0.0
Gly	0.2	0.0	0.1	0.1	0.0	0.0	0.1	99.4

Probability (in %) that Asp will mutated to Glu at evolutionary distance 1 PAM

Highest scores at the diagonal

Values in each column sum to 100%

Matrix is not strictly symmetric

$$m(x \rightarrow y) \neq m(y \rightarrow x)$$

however

$$f(y) \times m(x \rightarrow y) = f(x) \times m(y \rightarrow x)$$

M.D. Dayhoff, R.M. Schwartz, and B.C. Orcut (1978), In *Atlas of Protein Sequences and Structures*, pp. 345 - 352

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Substitution matrices

PAM (Point Accepted Mutations) family

Extreme cases

PAM0

	Ala	Arg	Asn	Asp	Cys	Gln	Glu	Gly
Ala	100	0	0	0	0	0	0	0
Arg	0	100	0	0	0	0	0	0
Asn	0	0	100	0	0	0	0	0
Asp	0	0	0	100	0	0	0	0
Cys	0	0	0	0	100	0	0	0
Gln	0	0	0	0	0	100	0	0
Glu	0	0	0	0	0	0	100	0
Gly	0	0	0	0	0	0	0	100

No changes in amino acids

PAM ∞

	Ala	Arg	Asn	Asp	Cys	Gln	Glu	Gly
Ala	f(A)	f(A)	f(A)	f(A)	f(A)	f(A)	f(A)	f(A)
Arg	f(R)	f(R)	f(R)	f(R)	f(R)	f(R)	f(R)	f(R)
Asn	f(N)	f(N)	f(N)	f(N)	f(N)	f(N)	f(N)	f(N)
Asp	f(D)	f(D)	f(D)	f(D)	f(D)	f(D)	f(D)	f(D)
Cys	f(C)	f(C)	f(C)	f(C)	f(C)	f(C)	f(C)	f(C)
Gln	f(Q)	f(Q)	f(Q)	f(Q)	f(Q)	f(Q)	f(Q)	f(Q)
Glu	f(E)	f(E)	f(E)	f(E)	f(E)	f(E)	f(E)	f(E)
Gly	f(G)	f(G)	f(G)	f(G)	f(G)	f(G)	f(G)	f(G)

Frequencies of amino acids in the dataset

Equal probability to mutate to each residue

M.D. Dayhoff, R.M. Schwartz, and B.C. Orcut (1978), In *Atlas of Protein Sequences and Structures*, pp. 345 - 352

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Substitution matrices

PAM (Point Accepted Mutations) family

Higher-order PAM mutation matrices

PAM matrix for evolutionary distance 2 PAM, or PAM2

$$\mathbf{M}_2 = \mathbf{M}_1 \times \mathbf{M}_1 = \mathbf{M}_1^2$$

Matrix multiplication

$$m_2(x,y) = \sum_{z \in \{y\}} m_1(x,z) \times m_1(z,y)$$

PAM *matrix for an arbitrary evolutionary distance n, $\mathbf{M}_n$, or PAMn*

$$\mathbf{M}_n = \mathbf{M}_1^n$$

PAM1

	Ala	Arg	Asn	Asp	Cys	Gln	Glu	Gly
Ala	98.7	0.0	0.1	0.1	0.0	0.1	0.2	0.2
Arg	0.0	99.1	0.0	0.0	0.0	0.1	0.0	0.0
Asn	0.0	0.0	98.2	0.4	0.0	0.0	0.1	0.1
Asp	0.1	0.0	0.4	98.6	0.0	0.1	0.5	0.1
Cys	0.0	0.0	0.0	0.0	99.7	0.0	0.0	0.0
Gln	0.0	0.1	0.0	0.1	0.0	98.8	0.3	0.0
Glu	0.1	0.0	0.1	0.6	0.0	0.4	98.7	0.0
Gly	0.2	0.0	0.1	0.1	0.0	0.0	0.1	99.4

PAM250

	Ala	Arg	Asn	Asp	Cys	Gln	Glu	Gly
Ala	13	6	9	9	5	8	9	12
Arg	3	17	4	3	2	5	3	2
Asn	4	4	6	7	2	5	6	4
Asp	5	4	8	11	1	7	10	5
Cys	2	1	1	1	52	1	1	2
Gln	3	5	5	6	1	10	7	3
Glu	5	4	7	11	1	9	12	5
Gly	12	5	10	10	4	7	9	27

M.D. Dayhoff, R.M. Schwartz, and B.C. Orcut (1978), In *Atlas of Protein Sequences and Structures*, pp. 345 - 352

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Substitution matrices

PAM (Point Accepted Mutations) family

Log-odds scoring matrices

Observed probability

The odds ratio

$$R_n(x,y) = \frac{m_n(x,y)}{m_\infty(x,y)}$$

Random (background) probability

Log-odds matrix

$$\log_{10} \frac{m_{250}(x,y)}{f(y)}$$

PAM250 mutation matrix

	Ala	Arg	Asn	Asp	Cys	Gln	Glu	Gly
Ala	13	6	9	9	5	8	9	12
Arg	3	17	4	3	2	5	3	2
Asn	4	4	6	7	2	5	6	4
Asp	5	4	8	11	1	7	10	5
Cys	2	1	1	1	52	1	1	2
Gln	3	5	5	6	1	10	7	3
Glu	5	4	7	11	1	9	12	5
Gly	12	5	10	10	4	7	9	27

$$10 \times \log_{10} \frac{m_{250}(\text{Arg},\text{Ala})}{f(\text{Ala})} =$$

$$10 \times \log_{10} \frac{6\ (\%)}{8.7\ (\%)} \approx 10 \times \log_{10} 0.69 \approx -1.6 \approx -2$$

PAM250 scoring matrix

	Ala	Arg	Asn	Asp	Cys	Gln	Glu	Gly
Ala	2	-2	0	0	-2	0	0	1
Arg		6	0	-1	-4	1	-1	-3
Asn			2	2	-4	1	1	0
Asp				4	-5	2	3	1
Cys					12	-5	-5	-3
Gln						4	2	-1
Glu							4	0
Gly								5

M.D. Dayhoff, R.M. Schwartz, and B.C. Orcut (1978), In *Atlas of Protein Sequences and Stru*

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Substitution matrices

BLOSUM (BLOcks Substitution Matrix) family

Dataset: 500 groups of local alignments without gaps of distantly related protein sequences

BLOSUM scoring scheme

$$R_n(x,y) = 2 \times \log_2 \frac{f_n(x,y)}{f(x) \times f(y)}$$

Observed frequency of xy pair in the considered alignments

Back ground frequency of xy pair in any protein sequence

n is maximum sequence identity in the alignments considered

BLOSUM62 matrix

	Ala	Arg	Asn	Asp	Cys	Gln	Glu	Gly
Ala	4							
Arg	-1	5						
Asn	-2	0	6					
Asp	-2	-2	1	6				
Cys	0	-3	-3	-3	-9			
Gln	-1	1	0	0	-3	5		
Glu	-1	0	0	2	-4	2	5	
Gly	0	-2	0	-1	-3	-2	-2	6

S. Henikoff and J.G.Henikoff (1992), Am 5-10919

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Comparison of matrices

PAM (Point Accepted Mutations) family (Dayhoff, 1976)

based on global alignments of closely related proteins

counts number of changes between observed and predicted ancestor sequences

all other PAM matrices are extrapolated from PAM1 matrix

higher-number means larger evolutionary distance (less similar sequences)

BLOSUM (BLOck SUbstitution Matrix) family (Henikoff, 1992)

based on local alignments of distantly related proteins

counts amino acids pairs directly from alignments

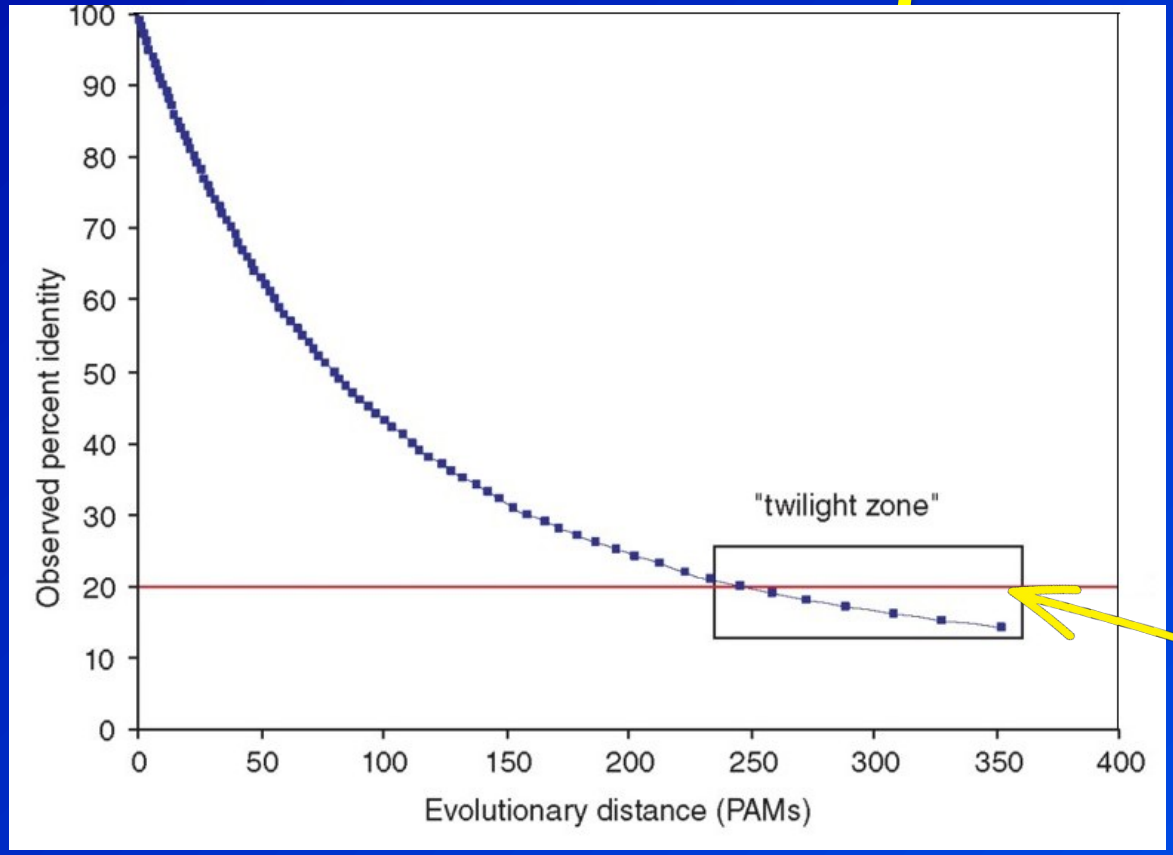
BLOSUM matrices are not extrapolated from each other

higher-number means smaller evolutionary distance (more similar sequences)

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Limits of homology detection

Percent identity



Proteins are homologous and easy to detect

Proteins may be homologous, but difficult to detect

Percent identity alone may be not sufficient to detect homology

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Statistical significance

Gaussian distribution

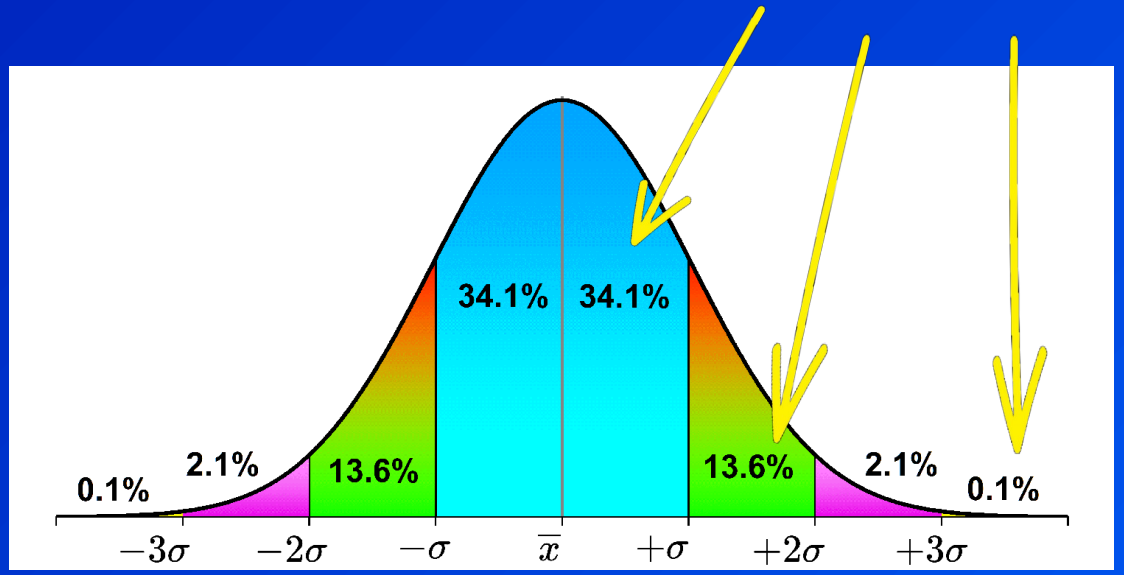
Fraction of the data within the interval

Mean for a sample of n data points ($x_1, x_2, \ldots, x_n$)

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Sample standard variance (deviation)

$$\sigma = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$



Z-score (standardized score) for global alignments

Generate large number of random alignments and calculate their scores

Calculate mean and sample standard deviation

Calculate Z-score of the original alignment

$$Z_i = \frac{x_{ali} - \bar{x}}{\sigma}$$

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Statistical significance

Problems with Z-score

Scores may not follow Gaussian distribution

Random alignments are not completely random → Non-Gaussian background distribution → Wrong estimation of probability by chance

Comparison of the sequence in question to large sequence databases

Large number of entries in the database → Elevated probability to find high score by chance → Significance level needs to be adjusted

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Statistical significance

Relative entropy

Definition for one aligned pair of residue and matrix S

$$H_S = \sum_{x,y} f_S(x,y) \times \log_2 \frac{f_S(x,y)}{f(x) \times f(y)}$$

Frequency of xy pair specific to a substitution matrix

Frequencies of residues x and y specific to a substitution matrix

Using relative entropy concept

PAM10 matrix

For one aligned pair H_{PAM10} = 3.43 bits → Minimum necessary entropy is 30 bits → Minimal length of significant alignment is 9 residues

PAM250 matrix

For one aligned pair H_{PAM250} = 0.36 bits → Minimum necessary entropy is 30 bits → Minimal length of significant alignment is 83 residues

Detection of remote homology requires longer sequence alignments

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Pairwise alignments

Prospectives

One of the most fundamental operations in bioinformatics

Active on-going assessment of available pairwise alignment algorithms

Further development of algorithms for new areas

Pitfalls

Optional algorithm parameters can greatly influence the outcome

Any two sequences, in principle, can be aligned, thus, additional information is needed to access biological relevance of the alignment

Percent identity does not reflect evolutionary changes

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Multiple Sequence Alignment

Identification of conserved regions and functional domains

Pfam, SMART, InterPro

Protein structure prediction

Co-evolution and contact prediction

Conserved regions

Functional protein annotation

Evolutionary relationships (Phylogenetics)

Phylogenetic trees and evolutionary distances

Post-translational modifications

Disease mutation analysis

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Multiple Sequence Alignment

Main approaches to construct multiple sequence alignment

- Exact
- Progressive sequence alignment
- Iterative
- Consistency-based
- Structure-based

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Multiple Sequence Alignment

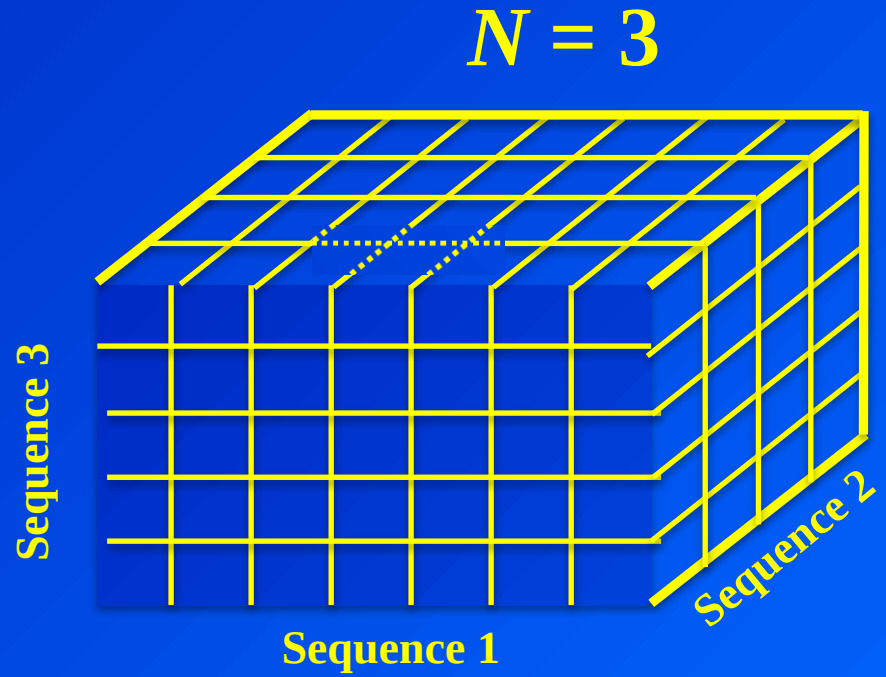
Exact approach

Extension of Dynamical Programming

N-dimensional dynamical programming matrix

Fill the matrix

Traceback



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Multiple Sequence Alignment

Exact approach

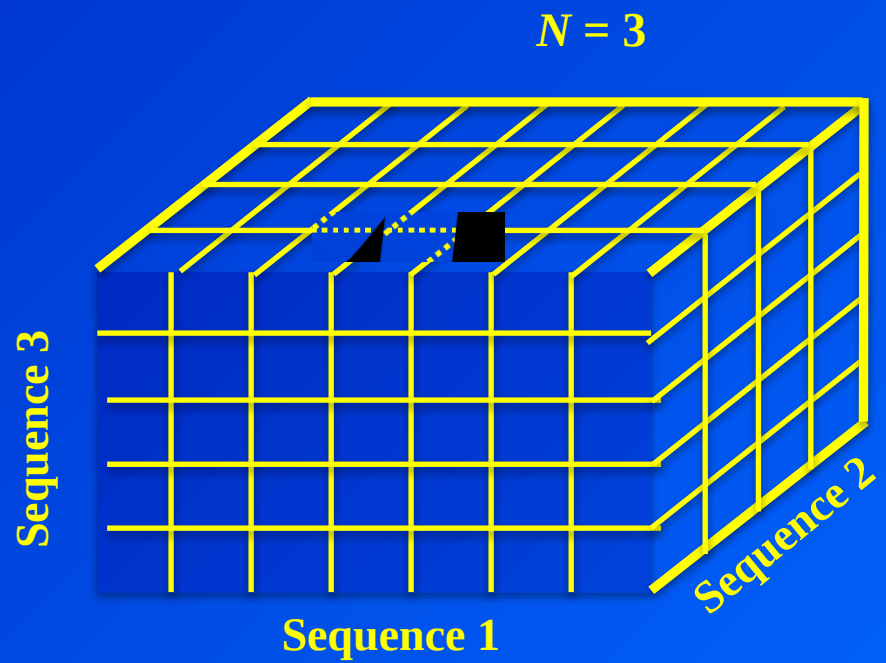
Unpractical for $N > 5$!

Extension of Dynamical Programming

N-dimensional dynamical programming matrix

Fill the matrix

Traceback



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Multiple Sequence Alignment

Progressive sequence alignment

Compute pairwise alignments and their scores

- Perform Needleman-Wunsch (Smith Waterman) alignments
- Construct a “distance” matrix

Construct guide tree based on scores of pairwise alignments

- Cluster sequences based on the distance matrix

Align sequences progressively

- Align the most closely related sequences first
- Merge them into profiles (aligned groups)
- Align profiles until all sequences are included.

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Multiple Sequence Alignment

Progressive sequence alignment

Exemplar sequences

```
Seq 1      MKTAVGLTDR
Seq 2      MKTGAVGLTD
Seq 3      MKVAAVGLTDR
Seq 4   MKTAGVGLT
Seq 5   MKTGAVGTD
Seq 6      MKAAAGLTDR
Seq 7   MKVGAVGLT
Seq 8   MKTAGVGTD
Seq 9      MKTAAGLTDR
Seq 10  MKVAAVGLADR
```

Percent similarity matrix, P_{ID}

Seq nr	1	2	3	4	5	6	7	8	9	10
1	100	82	82	73	50	80	64	50	90	73
2	82	100	73	70	90	64	80	64	73	64
3	82	73	100	73	73	82	73	64	82	91
4	73	70	73	100	56	55	67	78	64	55
5	50	90	73	56	100	50	67	78	60	46
6	80	64	82	55	50	100	55	50	90	73
7	64	80	73	67	67	55	100	44	55	64
8	50	64	64	78	78	50	44	100	60	46
9	90	73	82	64	60	90	55	60	100	73
10	73	64	91	55	46	73	64	46	73	100

```
BLOSUM62 NW-align
Seq 3      MKVAAVG TDR
Seq 10   MKVAAVGL DR
```

Number of identical residues

Length of the alignment

```
BLOSUM62 NW-align
Seq 7      MKV GA LT--
Seq 9      MKT AA  TDR
```

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Multiple Sequence Alignment

Progressive sequence alignment

Percent similarity matrix, P_{ID}

Seq nr	1	2	3	4	5	6	7	8	9	10
1	100	82	82	73	50	80	64	50	90	73
2	82	100	73	70	90	64	80	64	73	64
3	82	73	100	73	73	82	73	64	82	91
4	73	70	73	100	56	55	67	78	64	55
5	50	90	73	56	100	50	67	78	60	46
6	80	64	82	55	50	100	55	50	90	73
7	64	80	73	67	67	55	100	44	55	64
8	50	64	64	78	78	50	44	100	60	46
9	90	73	82	64	60	90	55	60	100	73
10	73	64	91	55	46	73	64	46	73	100

Seq 1 MKTAVGLTDR
Seq 2 MKTGAVGLTD
Seq 3 MKVAAVGLTDR
Seq 4 MKTAGVGLT
Seq 5 MKTGAVGTD
Seq 6 MKAAAGLTDR
Seq 7 MKVGAVGLT
Seq 8 MKTAGVGTD
Seq 9 MKTAAGLTDR
Seq 10 MKVAAVGLADR

"Distance" matrix, D

Seq nr	1	2	3	4	5	6	7	8	9	10
1	0	18	18	27	50	20	36	50	10	27
2	18	0	27	30	10	36	20	36	27	36
3	18	27	0	27	27	18	27	36	18	9
4	27	30	27	0	44	45	33	22	36	45
5	50	10	27	44	0	50	33	22	40	54
6	20	16	18	45	50	0	45	50	10	27
7	36	20	27	33	33	45	0	56	45	36
8	50	36	36	22	22	50	56	0	40	54
9	10	27	18	36	40	10	45	40	0	27
10	27	36	9	45	54	27	36	54	27	0

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Multiple Sequence Alignment

Progressive sequence alignment

"Distance" matrix, D

Seq nr	1	2	3	4	5	6	7	8	9	10
1	0	18	18	27	50	20	36	50	10	27
2	18	0	27	30	10	36	20	36	27	36
3	18	27	0	27	27	18	27	36	18	9
4	27	30	27	0	44	45	33	22	36	45
5	50	10	27	44	0	50	33	22	40	54
6	20	16	18	45	50	0	45	50	10	27
7	36	20	27	33	33	45	0	56	45	36
8	50	36	36	22	22	50	56	0	40	54
9	10	27	18	36	40	10	45	40	0	27
10	27	36	9	45	54	27	36	54	27	0

9

9

Seq 1 MKTAVGLTDR
Seq 2 MKTGAVGLTD
Seq 3 MKVAAVGLTDR
Seq 4 MKTAGVGLT
Seq 5 MKTGAVGTD
Seq 6 MKAAAGLTDR
Seq 7 MKVGAVGLT
Seq 8 MKTAGVGTD
Seq 9 MKTAAGLTDR
Seq 10 MKVAAVGLADR

Guide tree by hierarchical clustering

Treat each sequence as its own cluster

Find the closest pair of clusters.

Merge these clusters into a new cluster..

Calculate distance between clusters

Repeat until only one cluster remains

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Multiple Sequence Alignment

Progressive sequence alignment

"Distance" matrix, D

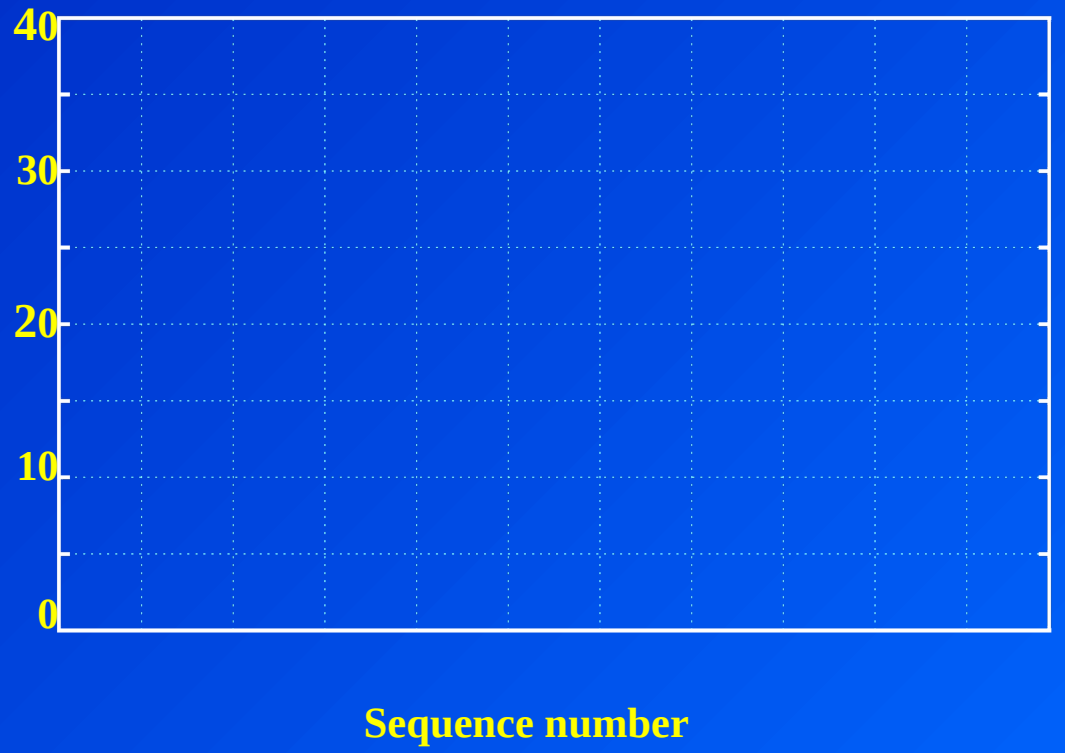
Seq nr	1	2	3	4	5	6	7	8	9	10
1	0	18	18	27	50	20	36	50	10	27
2	18	0	27	30	10	36	20	36	27	36
3	18	27	0	27	27	18	27	36	18	9
4	27	30	27	0	44	45	33	22	36	45
5	50	10	27	44	0	50	33	22	40	54
6	20	16	18	45	50	0	45	50	10	27
7	36	20	27	33	33	45	0	56	45	36
8	50	36	36	22	22	50	56	0	40	54
9	10	27	18	36	40	10	45	40	0	27
10	27	36	9	45	54	27	36	54	27	0

9

9

Seq 1 MKTAVGLTDR
Seq 2 MKTGAVGLTD
Seq 3 MKVAAVGLTDR
Seq 4 MKTAGVGLT
Seq 5 MKTGAVGTD
Seq 6 MKAAAGLTDR
Seq 7 MKVGAVGLT
Seq 8 MKTAGVGTD
Seq 9 MKTAAGLTDR
Seq 10 MKVAAVGLADR

Guide tree by hierarchical clustering



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Multiple Sequence Alignment

Progressive sequence alignment

"Distance" matrix, D

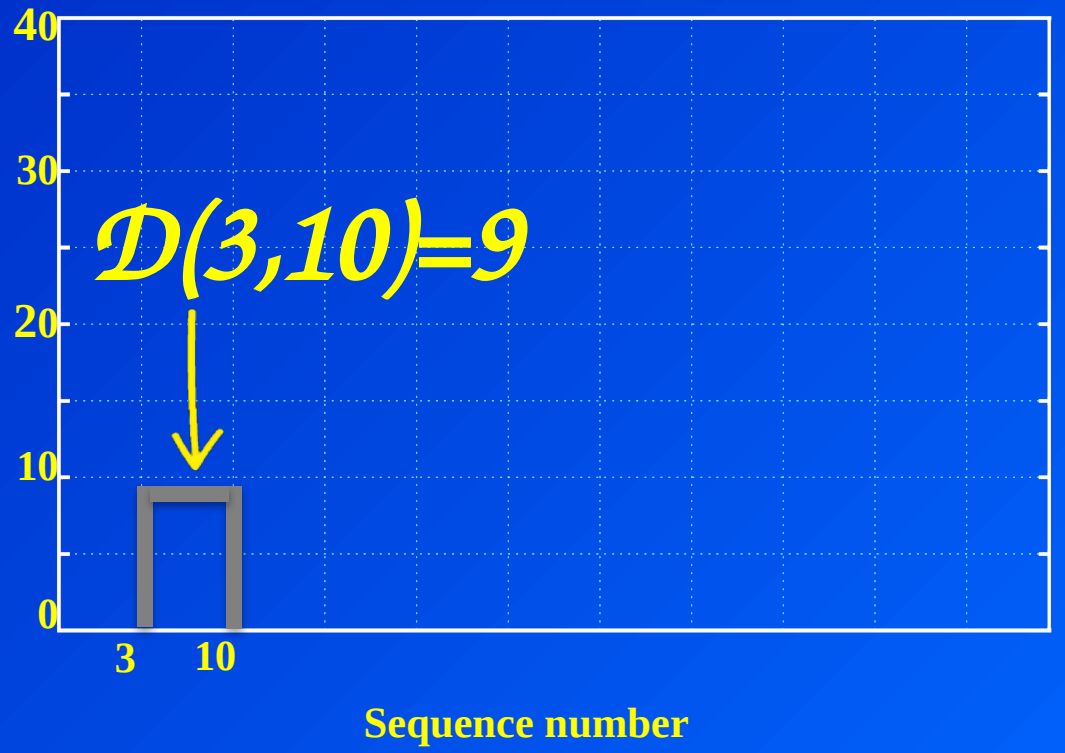
Seq nr	1	2	3	4	5	6	7	8	9	10
1	0	18	18	27	50	20	36	50	10	27
2	18	0	27	30	10	36	20	36	27	36
3	18	27	0	27	27	18	27	36	18	9
4	27	30	27	0	44	45	33	22	36	45
5	50	10	27	44	0	50	33	22	40	54
6	20	16	18	45	50	0	45	50	10	27
7	36	20	27	33	33	45	0	56	45	36
8	50	36	36	22	22	50	56	0	40	54
9	10	27	18	36	40	10	45	40	0	27
10	27	36	9	45	54	27	36	54	27	0

9

9

Seq 1 MKTAVGLTDR
Seq 2 MKTGAVGLTD
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Seq 4 MKTAGVGLT
Seq 5 MKTGAVGTD
Seq 6 MKAAAGLTDR
Seq 7 MKVGAVGLT
Seq 8 MKTAGVGTD
Seq 9 MKTAAGLTDR
Seq 10 MKVAAVGLADR

Guide tree by hierarchical clustering



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Multiple Sequence Alignment

Progressive sequence alignment

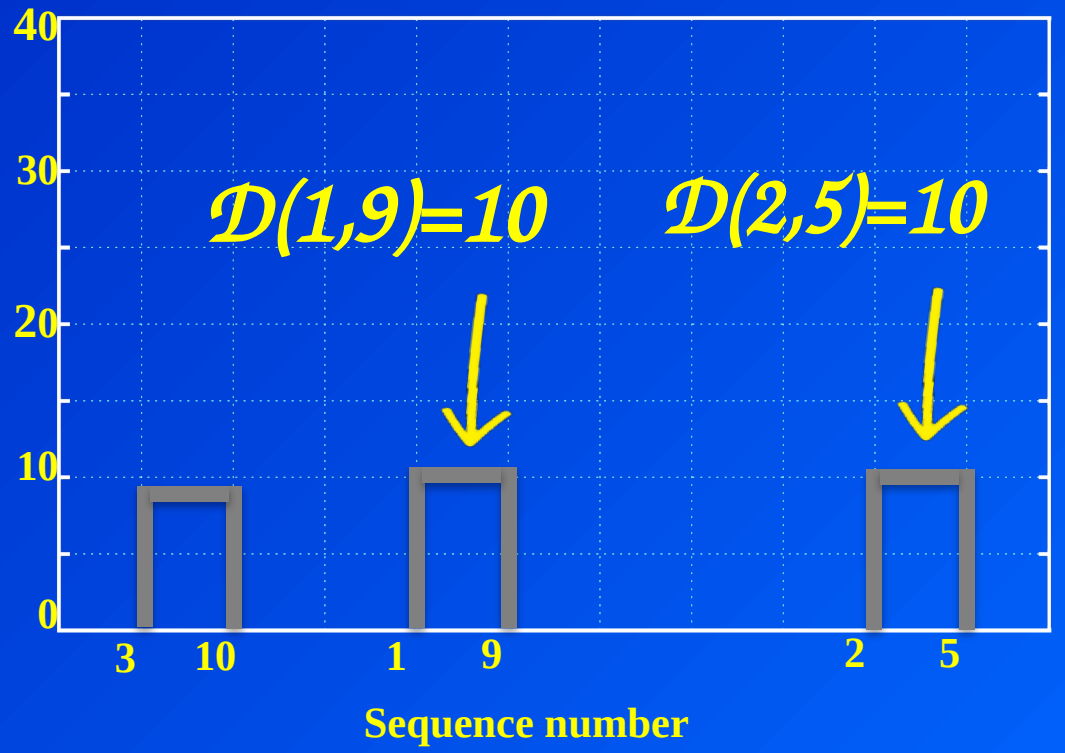
"Distance" matrix, D

Seq nr	1	2	3	4	5	6	7	8	9	10
1	0	18	18	27	50	20	36	50	10	27
2	18	0	27	30	10	36	20	36	27	36
3	18	27	0	27	27	18	27	36	18	9
4	27	30	27	0	44	45	33	22	36	45
5	50	10	27	44	0	50	33	22	40	54
6	20	16	18	45	50	0	45	50	10	27
7	36	20	27	33	33	45	0	56	45	36
8	50	36	36	22	22	50	56	0	40	54
9	10	27	18	36	40	10	45	40	0	27
10	27	36	9	45	54	27	36	54	27	0

10 10

Seq 1 MKTAVGLTDR
Seq 2 MKTGAVGLTD
Seq 3 MKVAAVGLTDR
Seq 4 MKTAGVGLT
Seq 5 MKTGAVGTD
Seq 6 MKAAAGLTDR
Seq 7 MKVGAVGLT
Seq 8 MKTAGVGTD
Seq 9 MKTAAGLTDR
Seq 10 MKVAAVGLADR

Guide tree by hierarchical clustering



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Multiple Sequence Alignment

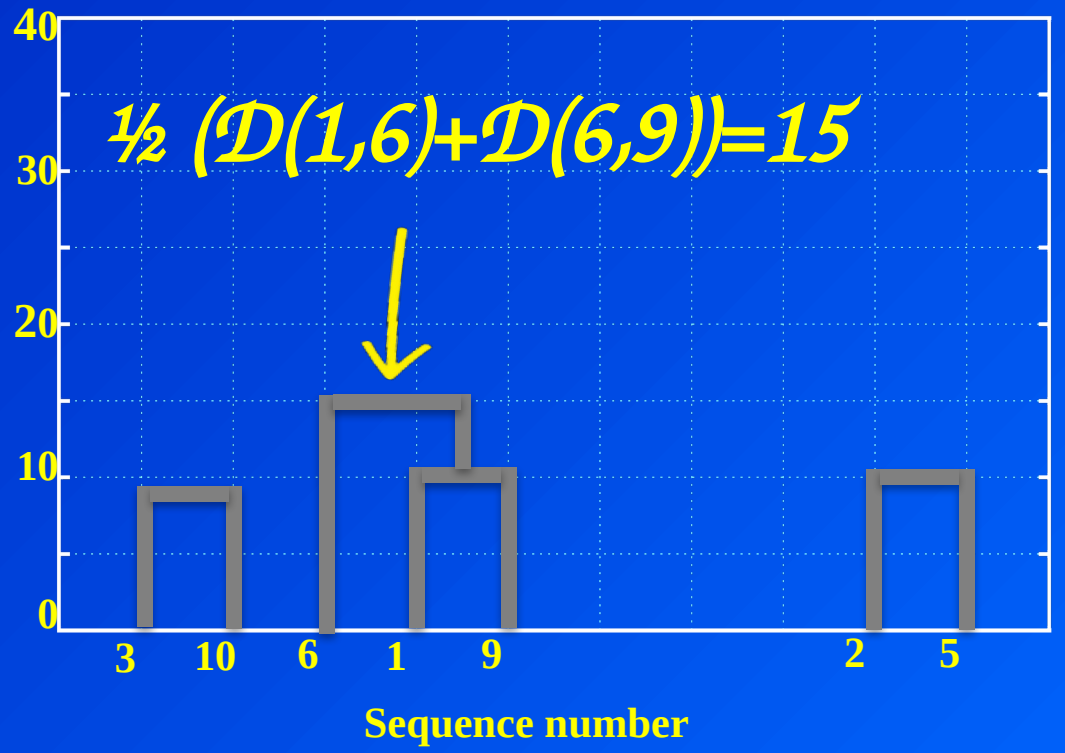
Progressive sequence alignment

"Distance" matrix, D

Seq nr	1	2	3	4	5	6	7	8	9	10
1	0	18	18	27	50	20	36	50	10	27
2	18	0	27	30	10	36	20	36	27	36
3	18	27	0	27	27	18	27	36	18	9
4	27	30	27	0	44	45	33	22	36	45
5	50	10	27	44	0	50	33	22	40	54
6	20	16	18	45	50	0	45	50	10	27
7	36	20	27	33	33	45	0	56	45	36
8	50	36	36	22	22	50	56	0	40	54
9	10	27	18	36	40	10	45	40	0	27
10	27	36	9	45	54	27	36	54	27	0

Seq 1 MKTAVGLTDR
Seq 2 MKTGAVGLTD
Seq 3 MKVAAVGLTDR
Seq 4 MKTAGVGLT
Seq 5 MKTGAVGTD
Seq 6 MKAAAGLTDR
Seq 7 MKVGAVGLT
Seq 8 MKTAGVGTD
Seq 9 MKTAAGLTDR
Seq 10 MKVAAVGLADR

Guide tree by hierarchical clustering



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Multiple Sequence Alignment

Progressive sequence alignment

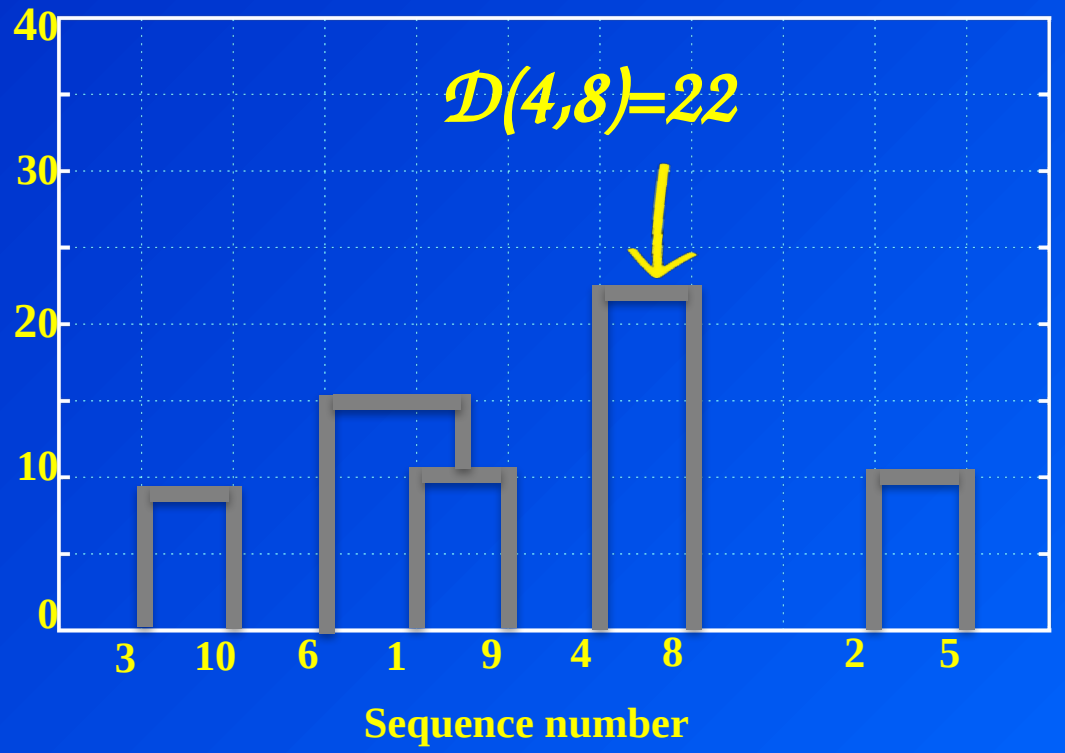
"Distance" matrix, D

Seq nr	1	2	3	4	5	6	7	8	9	10
1	0	18	18	27	50	20	36	50	10	27
2	18	0	27	30	10	36	20	36	27	36
3	18	27	0	27	27	18	27	36	18	9
4	27	30	27	0	44	45	33	22	36	45
5	50	10	27	44	0	50	33	22	40	54
6	20	16	18	45	50	0	45	50	10	27
7	36	20	27	33	33	45	0	56	45	36
8	50	36	36	22	22	50	56	0	40	54
9	10	27	18	36	40	10	45	40	0	27
10	27	36	9	45	54	27	36	54	27	0

22

Seq 1 MKTAVGLTDR
Seq 2 MKTGAVGLTD
Seq 3 MKVAAVGLTDR
Seq 4 MKTAGVGLT
Seq 5 MKTGAVGTD
Seq 6 MKAAAGLTDR
Seq 7 MKVGAVGLT
Seq 8 MKTAGVGTD
Seq 9 MKTAAGLTDR
Seq 10 MKVAAVGLADR

Guide tree by hierarchical clustering



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Multiple Sequence Alignment

Progressive sequence alignment

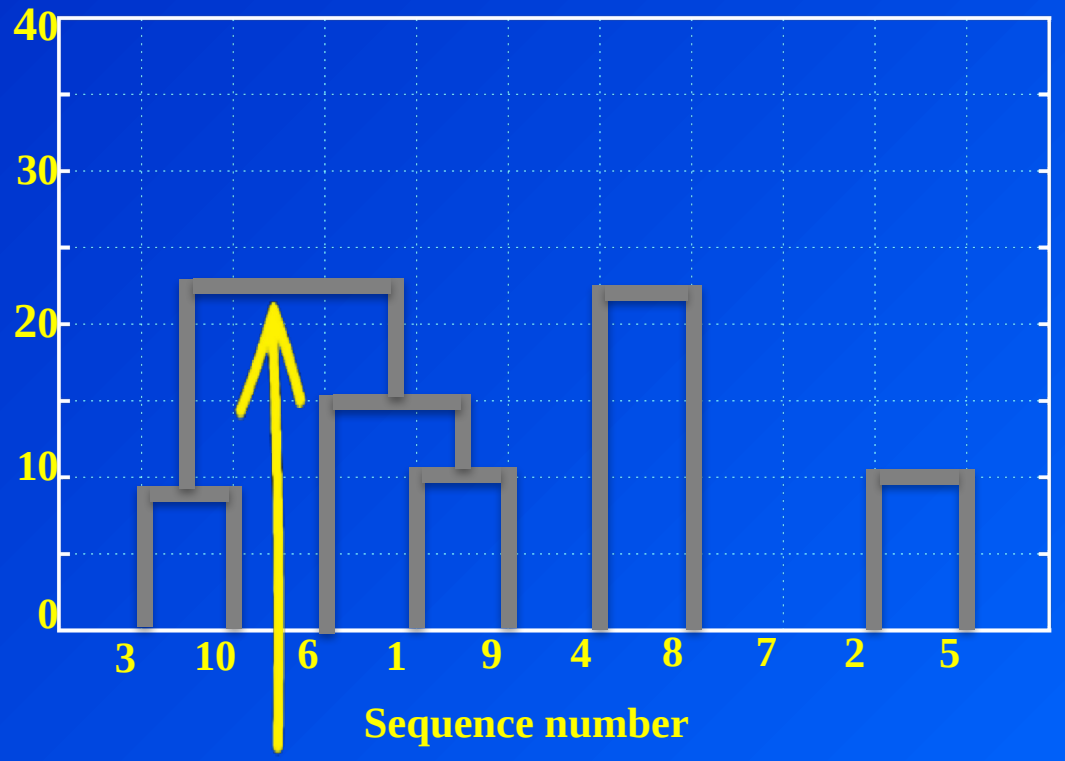
"Distance" matrix, D

Seq nr	1	2	3	4	5	6	7	8	9	10
1	0	18	18	27	50	20	36	50	10	27
2	18	0	27	30	10	36	20	36	27	36
3	18	27	0	27	27	18	27	36	18	9
4	27	30	27	0	44	45	33	22	36	45
5	50	10	27	44	0	50	33	22	40	54
6	20	16	18	45	50	0	45	50	10	27
7	36	20	27	33	33	45	0	56	45	36
8	50	36	36	22	22	50	56	0	40	54
9	10	27	18	36	40	10	45	40	0	27
10	27	36	9	45	54	27	36	54	27	0

18 18 18 27 27

Seq 1 MKTAVGLTDR
Seq 2 MKTGAVGLTD
Seq 3 MKVAAVGLTDR
Seq 4 MKTAGVGLT
Seq 5 MKTGAVGTD
Seq 6 MKAAAGLTDR
Seq 7 MKVGAVGLT
Seq 8 MKTAGVGTD
Seq 9 MKTAAGLTDR
Seq 10 MKVAAVGLADR

Guide tree by hierarchical clustering



$$1/6 \times (D(3,6)+D(3,1)+D(3,9)+D(10,6)+D(10,1)+D(10,9) = 22.5$$

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Multiple Sequence Alignment

Progressive sequence alignment

"Distance" matrix, D

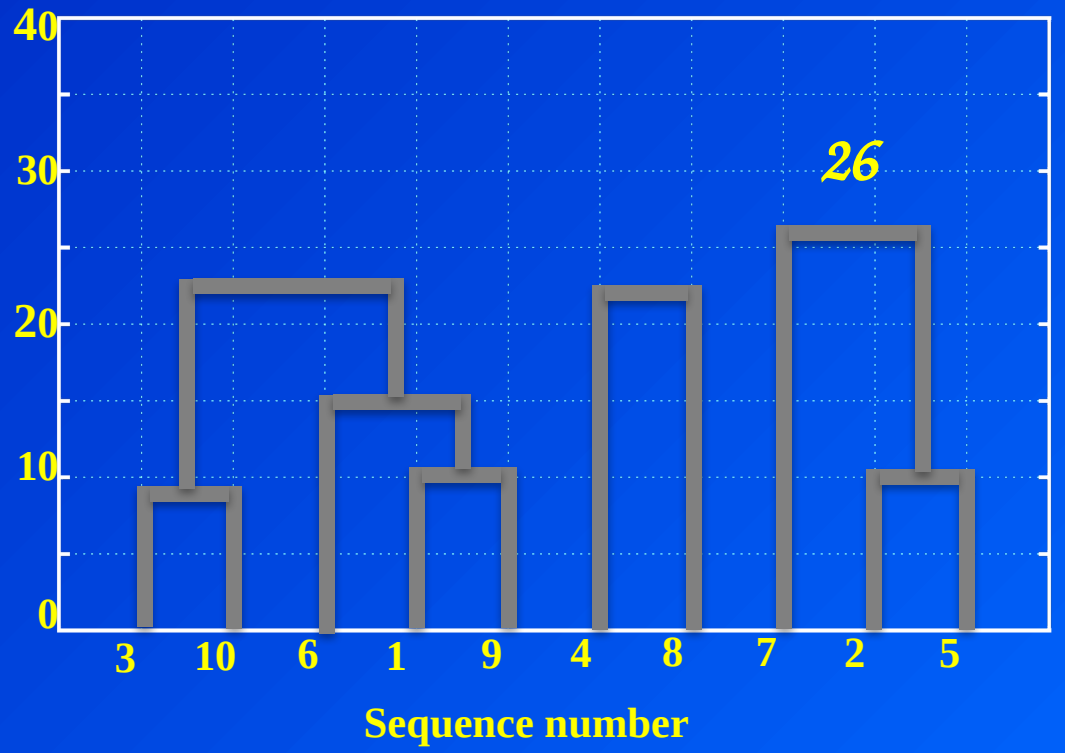
Seq nr	1	2	3	4	5	6	7	8	9	10
1	0	18	18	27	50	20	36	50	10	27
2	18	0	27	30	10	36	20	36	27	36
3	18	27	0	27	27	18	27	36	18	9
4	27	30	27	0	44	45	33	22	36	45
5	50	10	27	44	0	50	33	22	40	54
6	20	16	18	45	50	0	45	50	10	27
7	36	20	27	33	33	45	0	56	45	36
8	50	36	36	22	22	50	56	0	40	54
9	10	27	18	36	40	10	45	40	0	27
10	27	36	9	45	54	27	36	54	27	0

20

33

Seq 1 MKTAVGLTDR
Seq 2 MKTGAVGLTD
Seq 3 MKVAAVGLTDR
Seq 4 MKTAGVGLT
Seq 5 MKTGAVGTD
Seq 6 MKAAAGLTDR
Seq 7 MKVGAVGLT
Seq 8 MKTAGVGTD
Seq 9 MKTAAGLTDR
Seq 10 MKVAAVGLADR

Guide tree by hierarchical clustering



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Multiple Sequence Alignment

Progressive sequence alignment

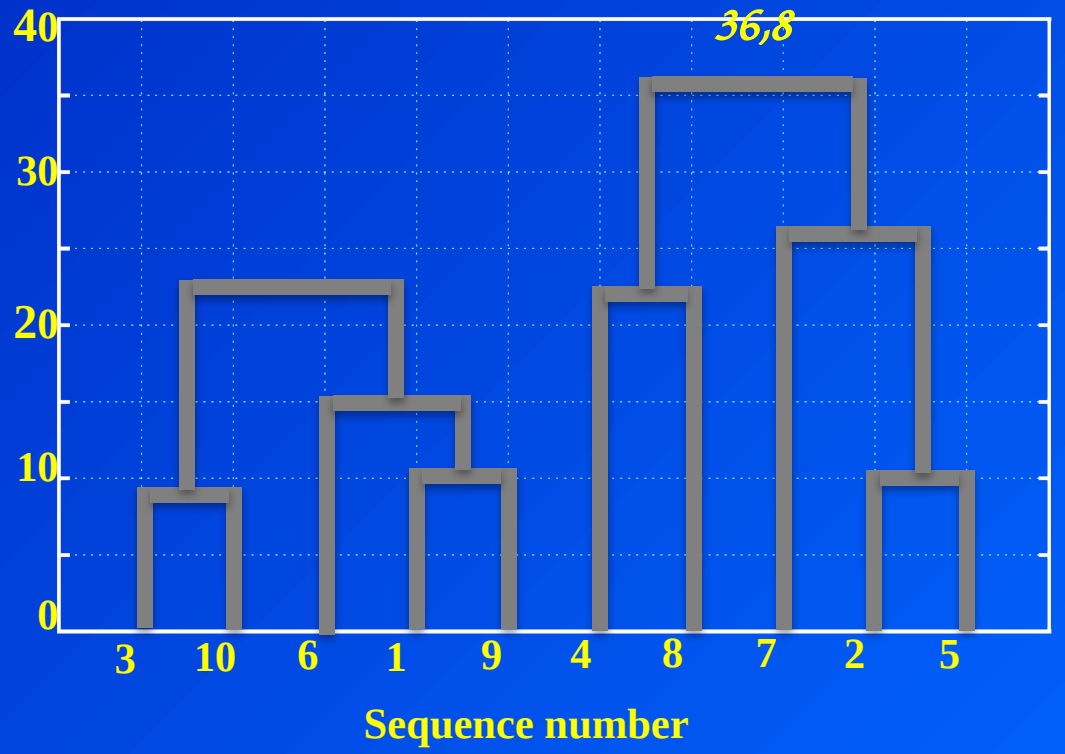
"Distance" matrix, D

Seq nr	1	2	3	4	5	6	7	8	9	10
1	0	18	18	27	50	20	36	50	10	27
2	18	0	27	30	10	36	20	36	27	36
3	18	27	0	27	27	18	27	36	18	9
4	27	30	27	0	44	45	33	22	36	45
5	50	10	27	44	0	50	33	22	40	54
6	20	16	18	45	50	0	45	50	10	27
7	36	20	27	33	33	45	0	56	45	36
8	50	36	36	22	22	50	56	0	40	54
9	10	27	18	36	40	10	45	40	0	27
10	27	36	9	45	54	27	36	54	27	0

30 44 33 36 22 56

Seq 1 MKTAVGLTDR
Seq 2 MKTGAVGLTD
Seq 3 MKVAAVGLTDR
Seq 4 MKTAGVGLT
Seq 5 MKTGAVGTD
Seq 6 MKAAAGLTDR
Seq 7 MKVGAVGLT
Seq 8 MKTAGVGTD
Seq 9 MKTAAGLTDR
Seq 10 MKVAAVGLADR

Guide tree by hierarchical clustering



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Multiple Sequence Alignment

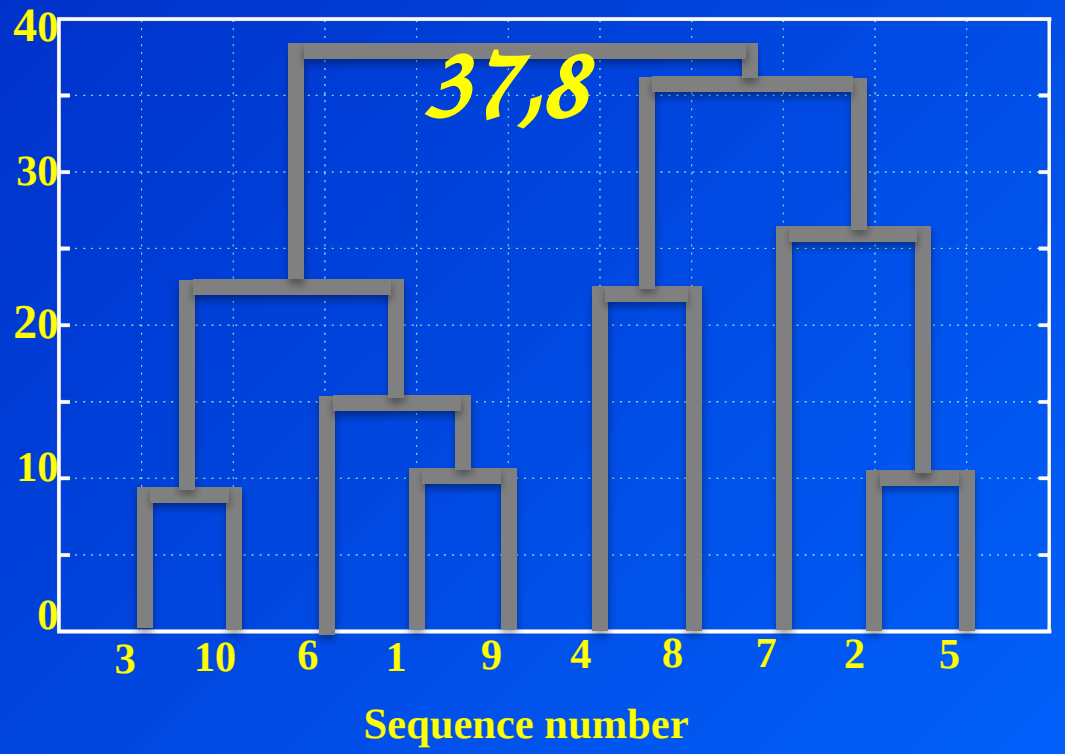
Progressive sequence alignment

"Distance" matrix, D

Seq nr	1	2	3	4	5	6	7	8	9	10
1	0	18	18	27	50	20	36	50	10	27
2	18	0	27	30	10	36	20	36	27	36
3	18	27	0	27	27	18	27	36	18	9
4	27	30	27	0	44	45	33	22	36	45
5	50	10	27	44	0	50	33	22	40	54
6	20	16	18	45	50	0	45	50	10	27
7	36	20	27	33	33	45	0	56	45	36
8	50	36	36	22	22	50	56	0	40	54
9	10	27	18	36	40	10	45	40	0	27
10	27	36	9	45	54	27	36	54	27	0

Seq 1 MKTAVGLTDR
Seq 2 MKTGAVGLTD
Seq 3 MKVAAVGLTDR
Seq 4 MKTAGVGLT
Seq 5 MKTGAVGTD
Seq 6 MKAAAGLTDR
Seq 7 MKVGAVGLT
Seq 8 MKTAGVGTD
Seq 9 MKTAAGLTDR
Seq 10 MKVAAVGLADR

Guide tree by hierarchical clustering

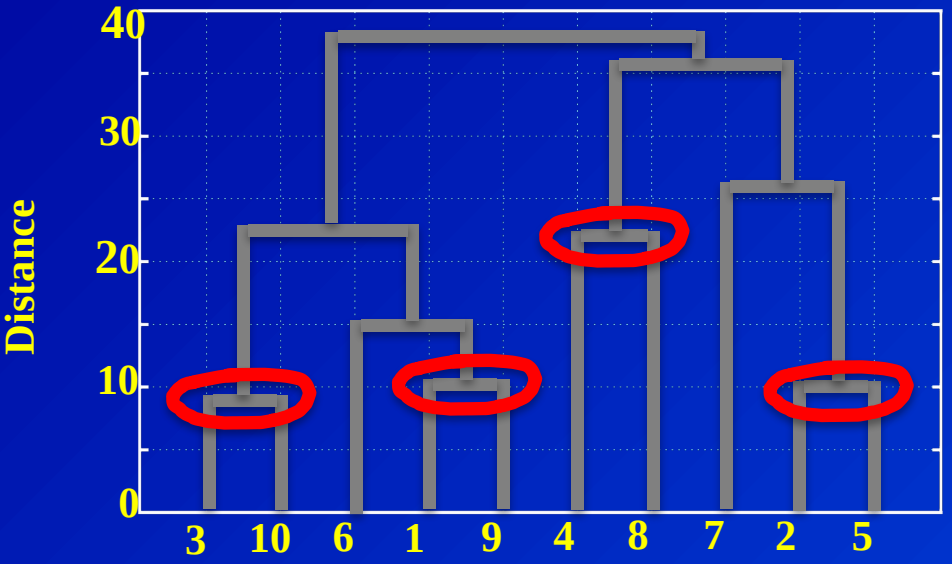


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Multiple Sequence Alignment

Progressive sequence alignment

Guide tree by hierarchical clustering




```
Seq 1     MKTAVGLTDR
Seq 2     MKTGAVGLTD
Seq 3     MKVAAVGLTDR
Seq 4   MKTAGVGLT
Seq 5   MKTGAVGTD
Seq 6     MKAAAGLTDR
Seq 7   MKVGAVGLT
Seq 8   MKTAGVGTD
Seq 9     MKTAAGLTDR
Seq 10  MKVAAVGLADR
```

Construct multiple sequence alignment

1st level (pairs)

```
Seq 3     MKVAAVGLTDR
Seq 10    MKVAAVGLADR

Seq 1     MKTAVGLTDR
Seq 9     MKTAAGLTDR

Seq 2     MKTGAVGLTD
Seq 5     MKTGAVG-TD

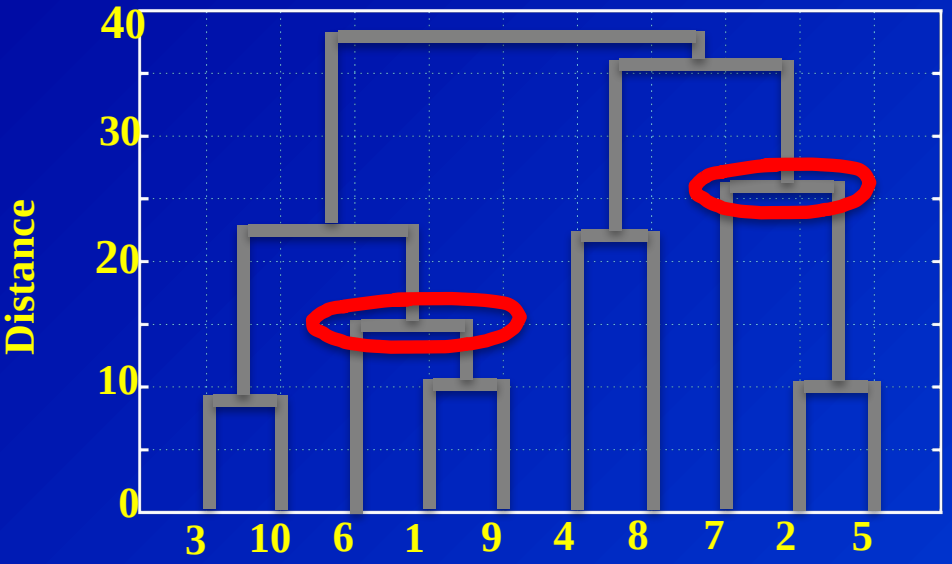
Seq 4   MKTAGVGLT
Seq 8   MKTAGVGTD
```

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Multiple Sequence Alignment

Progressive sequence alignment

Guide tree by hierarchical clustering




```
Seq 1     MKTAVGLTDR
Seq 2     MKTGAVGLTD
Seq 3     MKVAAVGLTDR
Seq 4   MKTAGVGLT
Seq 5   MKTGAVGTD
Seq 6     MKAAAGLTDR
Seq 7   MKVGAVGLT
Seq 8   MKTAGVGTD
Seq 9     MKTAAGLTDR
Seq 10  MKVAAVGLADR
```

Construct multiple sequence alignment

2nd level (triplets)

```
Seq 1     MKTAVGLTDR      Seq 2     MKTGAVGLTD
Seq 9     MKTAAGLTDR      Seq 5     MKTGAVG-TD
```

```
Seq 6     MKAAAGLTDR      Seq 7     MKVGAVGLT
```

```
Seq 6     MKAAAGLTDR      Seq 7     MKVGAVGLT-
Seq 1     MKTAVGLTDR      Seq 2     MKTGAVGLTD
Seq 9     MKTAAGLTDR      Seq 5     MKTGAVG-TD
```

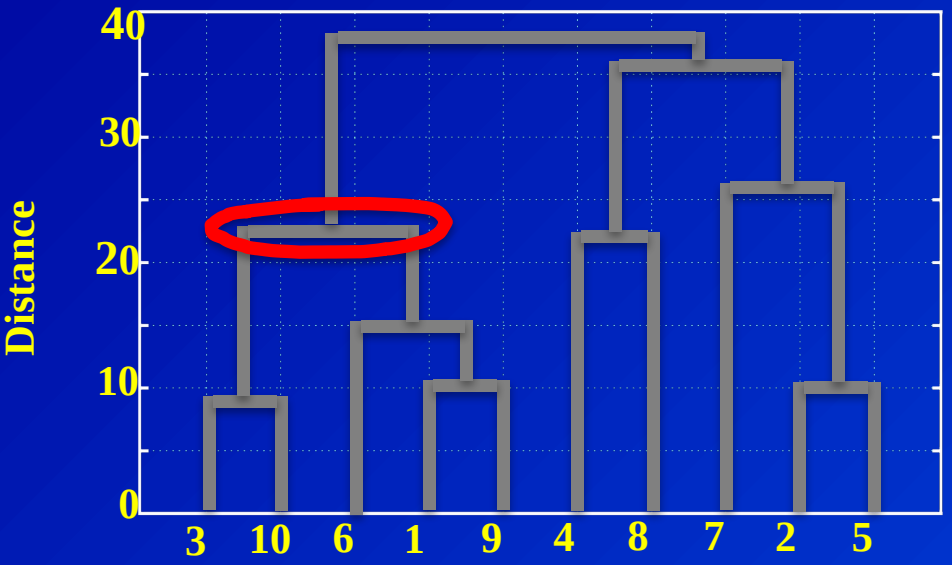
Gap from seq2 – seq7 pairwise alignment

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Multiple Sequence Alignment

Progressive sequence alignment

Guide tree by hierarchical clustering




```
Seq 1     MKTAVGLTDR
Seq 2     MKTGAVGLTD
Seq 3     MKVAAVGLTDR
Seq 4   MKTAGVGLT
Seq 5   MKTGAVGTD
Seq 6     MKAAAGLTDR
Seq 7   MKVGAVGLT
Seq 8   MKTAGVGTD
Seq 9     MKTAAGLTDR
Seq 10  MKVAAVGLADR
```

Construct multiple sequence alignment

3rd level (triplet + pair)

```
Seq 6     MKAAAGLTDR
Seq 1     MKTAVGLTDR
Seq 9     MKTAAGLTDR
```

+

```
Seq 3     MKVAAVGLTDR
Seq 10    MKVAAVGLADR
```

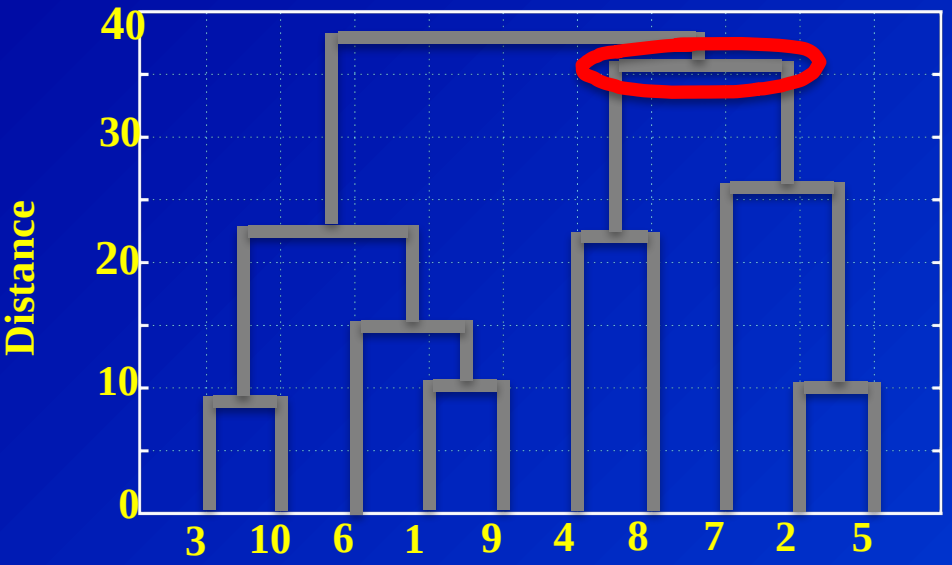
```
Seq 6     MKAAA-GLTDR
Seq 1     MKTA-VGLTDR
Seq 9     MKTAA-GLTDR
Seq 3     MKVAAVGLTDR
Seq 10    MKVAAVGLADR
```

Gap from pairwise alignments 39,13 and 36

Multiple Sequence Alignment

Progressive sequence alignment

Guide tree by hierarchical clustering




```
Seq 1     MKTAVGLTDR
Seq 2     MKTGAVGLTD
Seq 3     MKVAAVGLTDR
Seq 4   MKTAGVGLT
Seq 5   MKTGAVGTD
Seq 6     MKAAAGLTDR
Seq 7   MKVGAVGLT
Seq 8   MKTAGVGTD
Seq 9     MKTAAGLTDR
Seq 10  MKVAAVGLADR
```

Construct multiple sequence alignment

3rd level (triplet + pair)

```
Seq 7   MKVGAVGLT-
Seq 2   MKTGAVGLTD
Seq 5   MKTGAVG-TD
```

+

```
Seq 4   MKTAGVGLT
Seq 8   MKTAGVGTD
```

→

```
Seq 4   MKTAGVGLT-
Seq 8   MKTAGVG-TD
Seq 7   MKVGAVGLT-
Seq 2   MKTGAVGLTD
Seq 5   MKTGAVG-TD
```

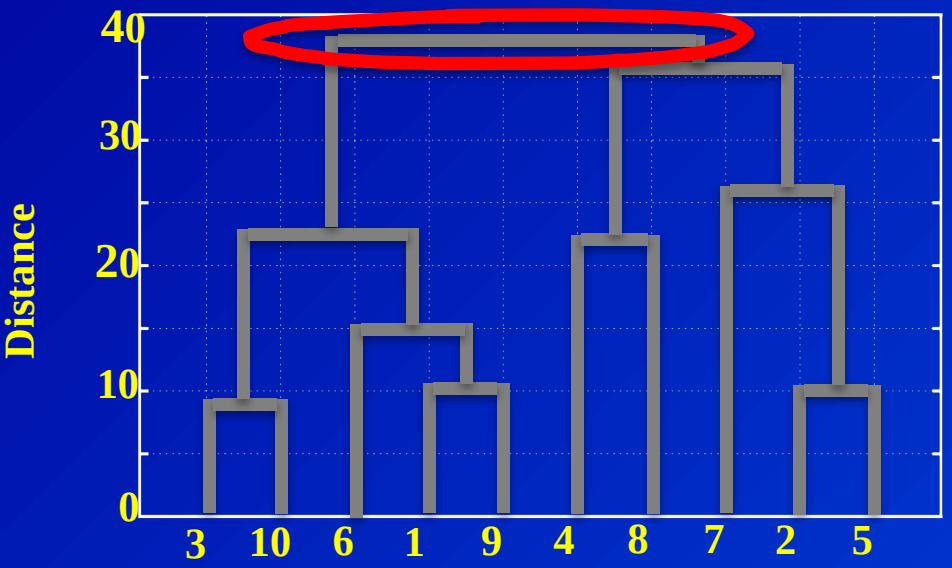
Gap from pairwise alignments 28 and 24

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Multiple Sequence Alignment

Progressive sequence alignment

Guide tree by hierarchical clustering




```
Seq 1     MKTAVGLTDR
Seq 2     MKTGAVGLTD
Seq 3     MKVAAVGLTDR
Seq 4   MKTAGVGLT
Seq 5   MKTGAVGTD
Seq 6     MKAAAGLTDR
Seq 7   MKVGAVGLT
Seq 8   MKTAGVGTD
Seq 9     MKTAAGLTDR
Seq 10  MKVAAVGLADR
```

Construct multiple sequence alignment

4th level (final MSA)

```
Seq 4     MKTAGVGLT-
Seq 8     MKTAGVG-TD
Seq 7     MKVGAVGLT-
Seq 2     MKTGAVGLTD
Seq 5     MKTGAVG-TD
```

+

```
Seq 6     MKAAA-GLTDR
Seq 1     MKTA-VGLTDR
Seq 9     MKTAA-GLTDR
Seq 3     MKVAAVGLTDR
Seq 10  MKVAAVGLADR
```

```
Seq 4     MKTAGVGLT--
Seq 8     MKTAGVG-TD-
Seq 7     MKVGAVGLT--
Seq 2     MKTGAVGLTD-
Seq 5     MKTGAVG-TD-
Seq 6     MKAAA-GLTDR
Seq 1     MKTA-VGLTDR
Seq 9     MKTAA-GLTDR
Seq 3     MKVAAVGLTDR
Seq 10  MKVAAVGLADR
```

Gap from pairwise alignments 34, 38, 37, 23 and 35

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Multiple Sequence Alignment

Iterative approach

- **Generate an initial alignment**
  - Progressive sequence alignment
- **Identify misligned regions**
  - Evaluate alignment using a scoring function
- **Remove and realign subsets of sequences**
  - Pairwise re-alignment of specific sequences
  - Progressive re-alignment of subsets
  - Tree-based re-alignment
- **Repeat until convergence**

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Multiple Sequence Alignment

Consistency-based approach

Generate an initial pairwise alignments

Stored in alignment profile

Consistency scoring

Agreement with indirect alignments

Progressive alignment with consistency enforcement

Iterative refinement

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Multiple Sequence Alignment

Comparison of MSA construction methods

Method	Programs	Speed	Accuracy	Guide tree dependence	Refinement	Area of applicability
Progressive	ClustalW	Fast	Moderate	High	No	Large datasets
Iterative	MUSCLE	Medium	High	Moderate	Yes	High accuracy
Consistency	T-Coffee	Slow	Very high	Low	Yes	Balanced speed and accuracy

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Main topics covered

Pairwise sequence alignment

Scoring matrices

Multiple sequence alignments